



City information models for optimal EV charging and energy-resilient renaissance

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IN BRIEF

Considering heavy computational loads in city-scale information models and future interactive electrification-driven internet of energy (IoE), this study develops an agent-based modeling (ABM) platform for optimal planning and distributions of urban cross-sector energy networks with multiple agents (eg, EVs, charging stations and buildings). City-level optimal layout of public EVCSs has been provided with climate-adaptive resilience and parameter uncertainties (like EV traveling behaviors, decision-making on charging piles selection, daily routes, arrival time, number of trips and so on). Afterward, the developed platform is applied for optimal EVCS infrastructure planning with material-energy-information nexus among building, transportation, grid and information networks in Nansha, China. Results show that the developed ABM platform is computationally efficient and accurate for energy-resilient city renaissance. This study first reports an agent-based city information model with optimal EV charging infrastructures for energy-resilient IoE-city renaissance.

City information models for optimal EV charging and energy-resilient renaissance

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BROADER CONTEXT

Electric vehicles (EVs) are being increasingly adopted, particularly in China, Europe, and the United States, as they offer a cleaner alternative to conventional vehicles. The integration of EVs with buildings and power grids establishes an internet of energy, facilitating multi-directional energy flows from distributed renewable energy systems. Central to this integration are electric vehicle charging stations (EVCSs), which enhance renewable energy penetration by connecting various components. Optimal planning of EVCSs is vital to determine locations and power levels, promoting efficient energy interactions. Considering challenges in cross-sector energy network modeling and future electrification-driven energy interactions, this study develops an agent-based modeling (ABM)-based city information model for optimal planning of urban public charging networks with multiple agents (eg, EVs, EVCSs, buildings, and the grid), as well as combining climate-adaptive resilience and EV behavior uncertainties for sustainable future cities.

ABSTRACT

Internet of energy (IoE) with building electrification provides multi-directional power interactions between E-mobility and power grid for smart and low-carbon transformations. However, the lack of a city-scale information model for planning infrastructures like electric vehicle charging stations leads to insufficient renewable self-consumption and performance overestimations. Furthermore, city-scale models struggle with achieving high accuracy for material-energy-information nexus across thousands of buildings, vehicles, and infrastructures. Considering future interactive energy dynamics, an agent-based modeling (ABM) platform is developed for optimal urban cross-sector energy network planning, incorporating climate-adaptive resilience and parameter uncertainties such as electric vehicle traveling behaviors, charging preferences, etc. Applied in Nansha, China, the ABM platform demonstrates computational efficiency and accuracy. It enhances urban renewable energy penetration by up to 50.1%, reduces road traffic carbon emissions by up to 46.6%, and decreases power outage hours and simulation time. This study offers a city-scale IoE model for optimal infrastructure planning and climate-adaptive energy-resilient renaissance.

INTRODUCTION

Global average temperature increase under the Paris Agreement needs to be kept well below 2°C above pre-industrial levels, with a concerted effort to limit the rise to 1.5°C.¹ Given that over 15% and 34% of carbon emissions are from road transport² and buildings,³ substantial decarbonization potentials from clean energy utilization can be achieved in road transport and buildings. In 2023, global electric vehicle (EV) sales approached 14 million, representing nearly 20% of total sales, with 95% of these sales concentrated in China, Europe, and the United States.² Electrification enables the interconnections and multi-interactions among buildings, transportation, and power grid, forming an interactive internet of energy. Specifically, in road transport, by replacing the traditional internal combustion engine vehicles, EVs can effectively reduce carbon emissions^{4–6} with improved environmental quality.⁷ From the building sector, EVs can work as distributed mobile energy storages for spatiotemporal energy sharing^{4,8} and as flexible loads to smartly adjust demand profiles.⁹ Furthermore, from the perspective of the power grid, EVs can help flatten grid power through Vehicle-to-Grid interactions.¹⁰ In addition, as the EV market share continues to increase, decarbonization potentials of EVs within the urban interactive energy network can be further improved.

Along with the electrification in vehicle markets, the development, deployment, and management of electric vehicle charging stations (EVCSs) become essentially important.^{11,12} As an intermediate bridge to connect both EVs and power grid, spatial distribution and power allo-

cation of EVCSs play essential roles in EV cruise anxiety and grid power adjustment/frequency regulation.¹³ Moreover, within the interactive internet of energy, EVCSs can simultaneously associate distributed renewables, buildings, EVs, and power grid for flexible power dispatch.⁴ To facilitate the electrification of vehicle industries, China government and industry provide increasing investments in public EVCSs in recent years.¹⁴ Public EVCSs include accessible infrastructures in various locations (such as residential buildings, offices, hotels, commercial buildings, schools, and so on) with commercial power trading operation among investors, operators, and EV owners based on total charged energy and peak power.^{11,15,16} Depending on different charging powers and speeds, public charging piles at EVCSs are commonly categorized into three levels, ie, Level 1 (slow), Level 2 (medium), and Level 3 (DC fast chargers).¹⁷ Instead of only charging EVs at home during nighttime at private piles, public EVCSs can enable longer cruises with energy accessibility.¹⁸ Optimal planning and distribution of public EVCSs within the city (including the location, power level, quantity of charging piles, etc.) are essential in improving renewable penetrations, energy flexibility for load dispatch, energy resilience for climate adaptation, and promoting EV market adoptions. However, the current academia fails to exploit EVCSs' functions within the interactive internet of energy in a future smart city, consisting of distributed renewables, buildings, EVs, power grids, and associated infrastructures. Furthermore, along with the interactive internet of energy transformation in a future smart city, the City Information Model (CIM) and techno-economic-environmental performance assessment platform have not been provided.

Over the past several years, studies on EVCS planning primarily emphasize the charging demand from the transportation side^{9,19} (like traveling demand coverage, charging service quality, and travel to CS cost), while ignoring renewable supply and penetration in EVs, buildings, and power grid throughout the EVCS. By tracking carbon emission sources, it is clear that electrification in transportation cannot actually achieve decarbonizations but move the emissions from vehicle side to the centralized coal-fired power plants, unless renewable energy accounts for a large share in energy supply.⁴ Deploying distributed renewable energy systems in buildings can significantly reduce carbon emissions in buildings, transportation, and power grid sectors.^{20–22} To better improve renewable penetrations and promote decarbonizations within the interactive internet of energy in a future smart city, several key processes need to be well designed, ie, renewable system energy for building prosumers,²³ surplus renewable-to-EV charging,²⁴ excess renewable exportation to the power grid,²⁵ and EV discharging for building demand coverages.^{8,26} However, current EVCS planning research solely concentrates on the transportation demand or the coupling between transportation and power grid sectors,^{9,27} while building prosumers with onsite renewable energy supply and demand of prosumers have not been considered for EVCS planning within the future city-scale internet of energy.²⁸

Sustainable and smart city renaissance requires a large-scale CIM, incorporating detailed information of thousands or even hundreds of thousands of buildings, vehicles, and infrastructures. Furthermore, information inputs, calculation, output result analysis, and decision-making in the smart city impose high requirements and great challenges in super-computing with high accuracy. In academia, studies on the city-scale integrative building-transportation energy systems are insufficient for smart city transformation. On the one hand, from the building aspect, energy interactions between EVs and buildings through charging piles are studied based on a few buildings and dozens of EVs at most.²⁹ On the other hand, for the public EVCSs planning, the already existing mathematical programming³⁰ models (eg, node-based models to cover the geographical area and flow-based models to maximize the vehicle flow with high utilization of charging piles³¹) can handle only a few thousand trips or vehicles at most, which is far below the hundreds of thousands or even millions of trips or vehicles routinely operating in large cities.³² In contrast, due to the superior capabilities in capturing complex and precise characteristics of micro-agents (like EV types, charging types,³³ modalities,³⁴ and grid integration³⁵), as well as fast response and efficient interactions among various agents at the city level, ABM enables fast calculation, super-computing, and decision-making in the large-scale city, together with overall performance evaluations of different energy management strategies and policies. However, none of these existing tools have considered dynamic renewable energy generations and demand patterns of building prosumers, incorporating multi-directional energy interactions and information exchange at the city scale.

Multi-energy systems in the smart city are full of uncertainties (like climate change with perturbation of solar radiation and wind³⁶ and EV behaviors), especially during the operation stage. Moreover, extreme heat waves and extreme events (like wars) may destroy the normal system operation and significantly influence the technical, economic, and environmental performances. Energy resilience under extreme events is essential to survive the power system and ensure the power supply during power outage periods.³⁷ To improve the city-scale energy resilience, the optimal EVCS planning can serve as a medium for vehicle-to-building (V2B) by offering adequate bidirectional charging piles to recharge the building and enabling users' accessibility,³⁷ even during the power outage periods. Nonetheless, the impact of optimal EVCS planning and distribution on city-scale energy supply resilience has been rarely considered. In addition, due to data limitations and computational resource constraints, only limited uncertain variables of EV driving behaviors (like one or two of traffic demand,¹⁰ service availability,¹⁶ trip characteristics,³⁸ energy supply,⁴ and charging pattern³⁹) have been considered, leading to the ineffectiveness of uncertainty analysis due to incomplete parameter considerations. Other uncertain parameters related to EV behaviors,

including the brand of a vehicle, the number of trips per day, the destination and time of each trip, driving speed, etc. have been considered in this research. In a word, the impact of optimal EVCS planning and distribution on city-scale energy resilience and robustness has not been provided considering parameter uncertainties and climate adaptation.

Based on the above review, research gaps can be summarized below.

- (1) The current research on city-scale public EVCS planning only focuses on the transportation demand coverage with grid power charging, while the electrified internet of energy with onsite renewable supply and demand of building prosumers has not been considered due to the complexity of energy system modeling and calculation for urban buildings. As active prosumers within the future smart city, multiple functions and flexible multi-agent access of building prosumers will essentially affect the public EVCS distributions, renewable penetrations, and city-scale decarbonizations.
- (2) Sustainable and smart city transformation imposes high requirements on the large-scale CIM, incorporating detailed information of thousands or even hundreds of thousands of buildings, vehicles, and infrastructures. Furthermore, information inputs, calculation, output result analysis, and decision-making in the smart city impose high requirements and great challenges in super-computing with high accuracy. However, the already existing model only involves a few buildings, a dozen vehicles, and a few thousand trips, which are insufficient for city-scale sustainability transformations.
- (3) Impact of optimal city-scale public EVCS planning and distribution on renewable energy penetration and carbon mitigation has not been quantified. Future climate change and extreme events will destroy the power supply of existing urban energy systems with power shortfall or power outage. Roles of energy infrastructures on city-scale energy resilience improvement are critical and essential to ensure the power supply reliability, even when the system suffers from extreme heat waves or extreme events (like wars). However, the impact of city-scale optimal EVCS planning on climate-adaptive energy resilience and robustness with EV behavior uncertainties has not been conducted.

To address the above-mentioned research gaps, research originalities are summarized below.

- (1) For internet of energy-based smart city renaissance, a CIM with optimal city-scale public EVCS planning and distribution has been developed, comprehensively considering onsite renewable supply and demand of building prosumers, transportation demands, and dynamic grid power. Transformation toward interactive electrification-driven internet of energy with multiple functions (like onsite clean energy supply and EV charging) and flexible multi-agent access of building prosumers has been studied with public EVCS distributions, renewable penetrations, and city-scale decarbonizations.
- (2) Regarding heavy interactional computational loads in city-scale information models with material-energy-information nexus of thousands of buildings, vehicles, and infrastructures, an ABM platform via super-computing in high accuracy has been developed for optimal planning and distributions on urban cross-sector energy networks, through transient performance simulation to assist the decision-making in city-scale sustainability transformations. The developed ABM platform can flexibly integrate multiple agents' integrations (eg, EVs, charging stations, buildings, etc.), incorporating detailed information of thousands or even hundreds of thousands of buildings, vehicles, and infrastructures.
- (3) In order to adapt to extreme heat waves or extreme events (like wars) with power supply reliability, city-level optimal layout of public EVCS has been provided with climate-adaptive resilience and energy robustness with parameter uncertainties (like EV traveling behaviors, decision-making on charging piles selection, daily routes, arrival time, number of trips, and so on). The proposed optimal layout of energy infrastructures can guide technicians, engineers, and policymakers in EVCS with city-scale

Table 1. Scenarios design with distributed renewable energy systems, building-E-mobility-grid multi-network energy interactions, and charging station planning strategies in different years

Scenarios	Renewable configurations	Energy interactions	EVCS planning strategy	Year
S1	No	No	Reference strategy	Typical ^d
S2	Rooftop PV ^a , BIPV ^b	No	Reference strategy	Typical
S3	Rooftop PV, BIPV, WT ^c	No	Reference strategy	Typical
S4	Rooftop PV, BIPV, WT	Yes	Reference strategy	Typical
S5	Rooftop PV, BIPV, WT	Yes	Renewable density strategy	Typical
S6	Rooftop PV, BIPV, WT	Yes	Traffic flow strategy	Typical
S7	Rooftop PV, BIPV, WT	Yes	Mix strategy	Typical
S8	Rooftop PV, BIPV, WT	Yes	Reference strategy	2020 ^e
S9	Rooftop PV, BIPV, WT	Yes	Climate change adaptation strategy	2020
S10	Rooftop PV, BIPV, WT	Yes	Reference strategy	2060 ^f
S11	Rooftop PV, BIPV, WT	Yes	Climate change adaptation strategy	2060
S12	Rooftop PV, BIPV, WT	Yes	Reference strategy	2100 ^g
S13	Rooftop PV, BIPV, WT	Yes	Climate change adaptation strategy	2100

^aThe PV denotes photovoltaic system.

^bThe BIPV denotes building-integrated PV system.

^cThe WT denotes wind turbine.

^dThe Typical refers to the use of a typical meteorological year (TMY) weather file exported from the professional software Meteonorm (see Table S1).

^eYear 2020 refers to the use of a 2020 weather file exported from the professional software Meteonorm (see Table S1).

^fYear 2060 refers to the use of a prediction weather file of 2060 exported from the professional software Meteonorm (see Table S1).

^gYear 2100 refers to the use of a prediction weather file of 2100 exported from the professional software Meteonorm (see Table S1).

energy resilience improvement and power supply reliability, even when the system suffers from power shortfall or power outage. Impact of optimal city-scale public EVCS planning and distribution on renewable energy penetration and carbon mitigation has been quantified.

Contributions of this study are summarized as follows. We developed a CIM with super-computing efficiency in high accuracy for city-scale public EVCS planning and distribution, comprehensively considering material-energy-information nexus of distributed renewables, stochastic building energy demands, driving behaviors of E-mobility, and dynamic response of power grid. An ABM platform has been developed for urban cross-sector energy networks with multiple agents' integrations (eg, EVs, charging stations, and buildings), incorporating detailed information of thousands or even hundreds of thousands of buildings, vehicles, and infrastructures. The agent-based model is then applied as a digital tool in the Guangdong-Hong Kong-Macao Greater Bay Area for optimal EVCS site locations and techno-economic-environmental performance estimation, to help flourish EV market share, city renovation, and sustainability transformation. The model also incorporates various uncertainties to ensure the energy supply reliability and robustness, including climate change-induced temperature and solar radiation fluctuations, EV scheduling model (ie, the number of trips, destinations, speed, and arrival times), and the availability of charging piles. Furthermore, considering future climate change and extreme events, and power outage under extreme weather events, energy resilience from optimal EVCS distributions can be further enhanced with power system survival and sufficient power supply during power outage periods. The proposed EVCS planning strategies can comprehensively contribute to transportation-side decarbonization, charging infrastructure utilization increase, urban renewable energy penetration boosting, and climate change adaptation. The Monte Carlo simulation results further verified the robustness and reliability of the simulation results.

This study is organized as introduction, results, discussion, and methods. In following sections, studied areas include optimal locations and capacities of EVCSs with techno-economic-environmental evaluation, validation of performance robustness, impact of climate change on building energy demands, and influence of EVCS planning on power outages.

RESULTS

Optimal layout of public EVCSs at the city level

In this study, 13 scenarios have been proposed, studied, and compared, as listed in Table 1. Five types of EVCS planning strategies are tested and compared in the scenarios.

The reference strategy is the current layout of public charging stations in Nansha, China, including 192 charging stations, locations (WGS84), 2–30 charging piles in a station, and the power levels of 7 kW (slow), 24 kW (medium), and 90 kW (fast). Depending on different comparison purposes, there are a total of seven scenarios with reference strategies (S1, S2, S3, S4, S8, S10, and S12).

The new EVCS planning strategies include the renewable density strategy, traffic flow strategy, mix strategy, and climate change adaptation strategy. Aside from the mix strategy, all other three new EVCS planning strategies aim to establish 300 charging stations to broaden EVCS coverage. Each station is limited to a maximum of 15 charging piles to manage capacity, with power levels evenly distributed among 7 kW (slow), 24 kW (medium), and 90 kW (fast).

The renewable energy density strategy (S5) aims to enhance the penetration of renewable energy (RE) by strategically mapping surplus RE from buildings to nodes within the road network. From this mapping, the top 300 nodes with the highest surplus RE are selected for the installation of EVCSs. Each node can have up to 15 charging piles allocated, with the number of piles proportional to the node's surplus RE. The power levels of these piles are adjusted based on the types of connected buildings. Specifically, 90-kW charging piles, which comprise 30%–40% of the distribution, are typically associated with office and commercial buildings. In contrast, 7-kW piles are designated for hotels, schools, and residential buildings, with the remaining capacity allocated to 24-kW piles.

The traffic flow strategy (S6) is designed to improve system traffic performance and vehicle-to-building (V2B) capabilities by initially assuming the presence of a charging pile at every node in the road network. Through a year-long simulation, where EVs can freely select their charging stations, the strategy identifies the optimal 300 nodes based on the frequency of EV charging. Charging piles are then distributed proportionally across these nodes, with power levels evenly divided among 7-kW, 24-kW, and 90-kW options.

In the mixed strategy (S7), the spatial locations and power level distributions of EVCSs are derived from the union of the renewable energy density and traffic flow strategies, slightly increasing the number of stations to 320 and charging piles per station to 20 to accommodate both building surplus RE and traffic flow considerations.

The climate change adaptation strategy aims to enhance V2B interactions and reduce the duration of power outages in buildings. This strategy determines the placement and quantity of charging piles at EVCSs based on the power outage durations at each node. Using the TRNSYS models for modeling transient energy demand of various building types, a benchmark limitation for power outages is established and adjusted according to each building's area (see [Tables S1](#) and [S2](#) and [Note S1](#)). This threshold helps guide the installation of EVCS at the top 300 nodes, with the number of charging piles determined linearly based on the average duration of power outages. The final distribution of charging piles is set according to a probability distribution, with 30% allocated to 7-kW piles, 30% to 24-kW piles, and 40% to 90-kW piles.

Each strategy is meticulously designed to address specific challenges and opportunities within the EVCS deployment landscape, offering tailored solutions to enhance infrastructure readiness and sustainability.

Taking Nansha as an example, the urban energy internet connects building prosumers, electric road network, and the power grid through EVCSs. The urban infrastructure network configuration, energy flow, and optimal EVCS planning map are shown in [Figure 1](#). The information and their representative colors of five types of buildings are given in [Figure 1A](#). The building representative colors are consistent in all figures. In [Figure 1B](#), a community consisting of five types of buildings, roads, two EVCSs, and EVs is shown, along with energy flows. It is noted that all energy flows are through charging stations (like grid-to-vehicle [G2V], building-to-vehicle [B2V], or V2B), illustrating the role of EVCS as an energy interaction medium in the urban energy network.

In [Figure 1C](#), five EVCS strategies (including the reference strategy) exhibit unique spatial and capacity distribution characteristics. The renewable density strategy, the traffic flow strategy, and the mix strategy focus on the surplus RE and EV charging counts as given in [Figures S1A](#) and [S1C](#), which are common in residential buildings and commercial buildings. The climate change adaptation strategy emphasizes the risk of power outages and building energy shortages as given in [Figures S1B](#) and [S1D](#), which are heavy in office buildings, school buildings, and residential buildings. To promote the connections and energy interactions between EVCSs and buildings, compared with the reference strategy with large capacities in an individual charging station, the four proposed strategies have a greater number of EVCSs with limited capacities. Overall, [Figure 1](#) demonstrates the composition of the ABM-based urban energy internet, the micro-agents and their attributes of the ABM model, and the optimal EVCS planning results along with data basis. Afterward, the technical-economic-environmental evaluation and robustness test of the four proposed EVCS planning methods will be analyzed in detail.

Technical-economic-environmental performance assessment

The technical-economic-environmental evaluation results show that compared with the reference strategy, the proposed new EVCS planning strategies have advantages in terms of traffic charging demand coverage, decarbonization of both on the building side and transportation side, EVCS owner benefits, and reduced power outage hours.

Charging station planning-based multi-network energy interactions display prospects in both transportation-side and building-side carbon reduction and sustainable development. In [Figure 2A](#), scenarios S5, S6, and S7 show fewer missed trips and unfinished distance compared with S4, with missed trips falling from 39.96% in S4 to 7.43%–16.84% in S5–S7, and unfinished distance dropping from 30.78% in S4 to 8.24%–16.76% in S5–S7. Similar reductions are observed in 2020, 2060, and 2100 scenarios, with missed trips decreasing to 3.13%–4.05% (S9 and S13) and unfinished distance to 3.50%–4.54% (S9 and S13). These improvements are attributed to the presence of sufficient EVCSs near residential areas in S5–S7, S9, S11, S13, and the ability for EVs to fully

charge overnight. It is worth noting that although S1–S3 show the perfect with zero missed trips and zero unfinished distance without V2B interactions, the EV charging totally relies on coal-fired power, as shown in [Figure 2B](#). In contrast, in scenarios with energy interactions (S4–S13), the proportion of green electricity (ie, from RE in buildings) in EV power sources is consistently above 28.8%, with S5 reaching the highest proportion at 46.6% as well as low missed trips and miles simultaneously in [Figure 2A](#).

Furthermore, as EV owners focused on EV battery capacity degradations before letting EVs participate in multi-directional energy interactions, this section also quantifies EV battery capacity degradation percentage (see [Tables S3](#) and [S4](#) for degradation curve parameters). Results in [Figure 2C](#) indicate that EV battery degradation is relatively mild and could possibly be offset by energy transactions. Specifically, in a 1-year simulation, scenarios involving energy interactions (S4–S13) show EV battery capacities ranging from 0.7% (first quartile) to 1.9% (third quartile), slightly higher than scenarios without energy interactions (S1–S3) at 0.5%–0.9%.

The results in [Figure 2D](#) indicate that the charging piles in scenarios with new EVCS strategies (S5–S7, S9, S11, and S13) have longer daily working time than those using the reference strategy (S1–S4, S8, S10, and S12). Particularly, the violin bodies (ie, KDE curves) show an even distribution in the 0- to 15-h range, with the first quartile of box-plots notably larger at approximately 5 h daily for 7 kW, 24 kW, and 90 kW charging piles. The results in [Figures 2A–2D](#) demonstrate that the proposed EVCS strategies with energy interactions (V2B and B2V) can enhance decarbonization effects and extend the service lifetime and utilization efficiency of infrastructures of the road traffic system, while effectively meeting transportation demand with low impact on EV battery capacity.

Effective EVCS planning can enhance the impact of distributed building RE systems and EV-based energy interactions on boosting RE penetration both at the city-level and building-level (results at building-level, see [Figure S2](#)). In [Figure 2E](#), the proposed EVCS strategies demonstrate an improvement in energy performance. In a typical year, the demand coverage ratio (DCR) of S5 and S7 has increased by 0.19% and 0.12% compared with S4, with no significant changes in 2020, 2060, and 2100. A similar trend can also be observed in renewable penetration ratio (RPR). The marginal increase can be attributed to several factors. First, DCR heavily relies on the ratio of RE supply. Although the proposed strategies provide ample opportunities for V2B interactions, most EVs tend to charge fully at night near residential buildings, where there is low-price electricity but limited renewables available for EV charging. Second, RPR, which is influenced by the ratio of RE used for building and travel demand, benefits from the chances for B2V interactions from the proposed strategies. However, only a small percentage of EVs can be charged during the day when surplus RE is available. Last, there are no external dispatch or economic signaling instructions forcibly imposed on EVs to facilitate energy interactions.

In [Figure 2F](#), the Sankey diagram shows that in Nansha, the energy demand of EV driving is much smaller than that of buildings. In addition, the RE penetration of residential buildings, office buildings, and commercial buildings is slightly greater than 50%, while school buildings and hotels are basically powered by RE. This is because, in Nansha, the former (residential buildings, office buildings, and commercial buildings) are mostly high-rise buildings (ie, building energy consumption is much greater than RE supply) and the latter (school buildings and hotels) are low-rise buildings (ie, building energy consumption is relatively small). Furthermore, half of the RE is exported to the power grid to decarbonize the power grid. This is because the incidental energy transportation of EVs alone is not enough to achieve city-level zero carbon, which puts forward requirements for technologies such as EV fleet scheduling and energy trading. The results in [Figures 2E](#) and [2F](#) demonstrate that even without external dispatch or economic signals, strategic planning of EVCSs can contribute to the increased penetration of RE in cities and buildings through the incidental energy interaction behaviors of EVs.

In [Figure 2G](#), the left vertical axis reveals that the proposed strategies increase service profits of EVCS owners by extending charging pile

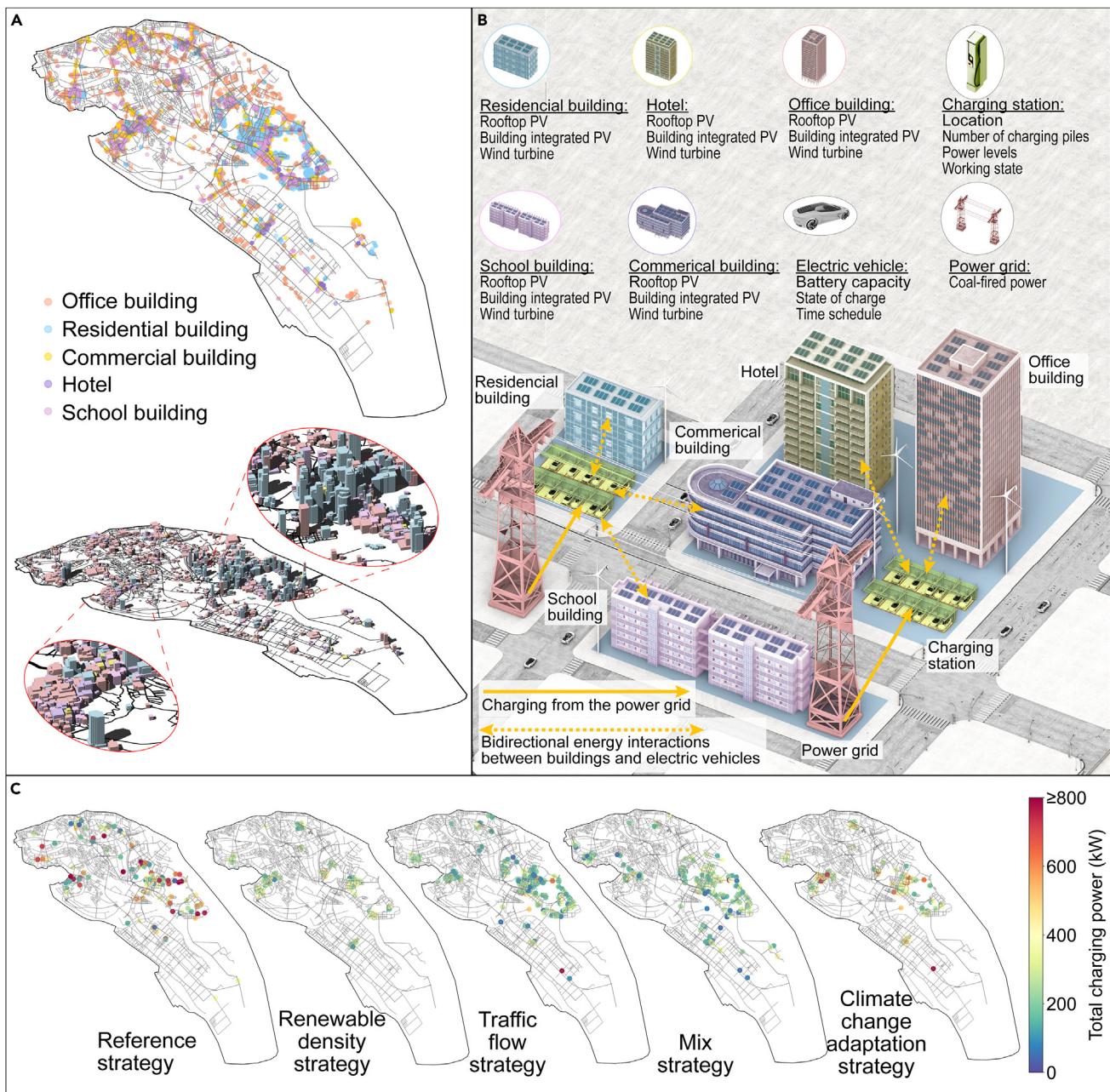


Figure 1. Internet of energy (IoE) of urban energy systems with energy and information interactions in Nansha (Guangdong), consisting of traffic network, distributed renewable energy network, EV charging infrastructure network, and the coal-fired power grid

(A) The exploded map shows the information of buildings including building types, locations, building heights, and building layout.

(B) The diagram shows community-scale energy interactions between buildings, charging stations, EVs, and the coal-fired power grid.

(C) Five EVCS location strategies (ie, reference strategy, renewable density strategy, traffic flow strategy, mix strategy, and climate change adaptation strategy) are shown on the traffic network in terms of locations and total charging power.

operating hours. Particularly, S4–S13 profits exceed S1–S3 by four orders of magnitude. In the typical year, EVCS service profits of S5 and S7 are 41% and 60% higher than that of S4. In 2020, 2060, and 2100, profits from climate change adaptive strategies rise by 14.2%, 16.7%, and 18.9%, respectively, vs. the reference strategy. This is due to the enhanced V2B and B2V interactions (Figure S2C). The right vertical axis shows that in the typical year, compared with S4 (reference strategy), the EVCS cost in S6 (traffic flow strategy) is 29% lower, while the EVCS costs in S5 (renewable density strategy) and S7 (mix strategy) are 8% and 15% higher. This advantage of S6 is due to S6 prioritizing EV charging over energy interactions, resulting in fewer charging piles than S4 with limited space demand. In

contrast, S5 emphasizes RE density and interactions, both of which are related to widespread buildings, leading to slightly more charging piles than S4. In the climate change adaptation strategy, EVCS costs decrease by 11%, with fewer charging piles addressing power outages and energy shortages. Overall, results in Figure 2G indicate economic feasibility of proposed EVCS strategies with around 10^{10} USD benefits of the EVCS service market for 1 year. Despite slightly higher costs for RE density strategy and traffic flow strategy, they yield the highest EVCS service profits for the 1-year simulation. All strategies show lower import costs and provide export profits and electricity bill savings with magnitude orders of 10^8 as additional benefits (see Figure S3).

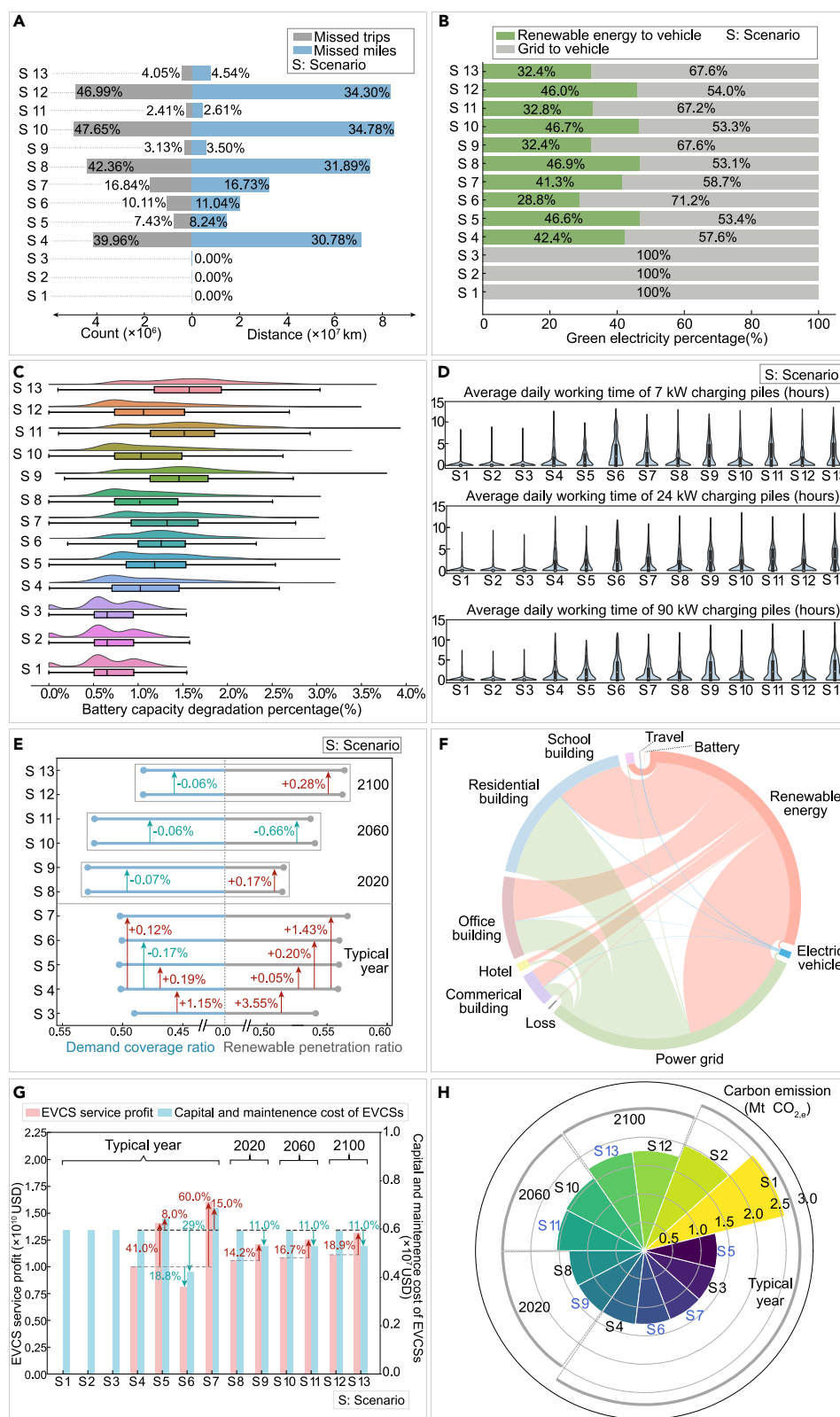


Figure 2. Traffic-side performance indicators for all scenarios

(A) Total number of missed trips and missed miles of all EVs in Nansha in each scenario, and the proportion of missed trips and missed miles in the total number of trips and trip distance, respectively. Missed trips represent the number of unfinished trips. A trip is considered unfinished if it does not start from the departure place or terminate at the destination. Missed miles mean the unfinished distance in a missed trip.

(B) Green electricity share in EV charging supply in each scenario. The green electricity comes from the surplus renewable energy of distributed renewable systems in buildings.

(legend continued on next page)

By smartly interconnecting distributed renewables, EVs, and power grid, effective EVCS planning can reduce carbon emissions in both urban buildings and transportation systems. In [Figure 2H](#), the EVCS strategies consistently show greater carbon emission reductions compared with the reference strategy across all years. Specifically, in a typical year, S5 (renewable density strategy) achieves the lowest carbon emissions at around 1.27 Mt CO_{2e}. This carbon emission reduction is attributed to the RE density strategy, which strategically plans EVCS locations and capacities based on building integration and the surplus RE from interconnected buildings at specific nodes in the road traffic network. This approach enhances B2V capabilities and RE integration, leading to reduced carbon emissions. The components of carbon emissions of all scenarios are given in [Figure S4](#). The findings in [Figure 2H](#) show that strategic planning of EV charging systems following the proposed strategies can aid in decarbonizing urban integrated building and transportation systems. The renewable density strategy (S5) based on the available surplus RE from buildings for EV charging can achieve the lowest total carbon emissions. Importantly, the climate change adaptation strategy with V2B interaction to reduce the number of power outage hours can successfully reduce emissions and tackle power outages, simultaneously.

Uncertainty analysis on optimal EVCS for performance robustness

Considering parameter uncertainties of the integrative urban energy systems, this section conducts uncertainty analysis via Monte Carlo simulations. The robustness of the integrated building-E-mobility network energy simulation platform is demonstrated through EV behaviors uncertainty modeling and Monte Carlo analysis. As shown in [Figure 3](#), the S5 is selected in this section for uncertainty analysis with 20 samples. [Figure 3A](#) shows the multivariate uncertainty modeling of EV behaviors based on uncertainty intervals and probability distribution extracted from the traveling behaviors survey report of Guangzhou, China (see [Tables S5–S11](#) for detailed information of uncertainty models).

In [Figure 3B](#), the Monte Carlo simulation results of traffic indicators are displayed. The fluctuation ranges of the five traffic indicators are acceptable. Specifically, based on the scatterplot and boxplot analysis, the missed trips fluctuate between 6.29% and 10.32%, and missed distance fluctuates between 6.91% and 10.92%, respectively. The EV charging green electricity percentage fluctuates between 46.30% and 47.18%. Although the maximum EV battery capacity degradation varies by about 0.9%, the distributions depicted by the kernel density estimation (KDE) curve and boxplot are very similar. This trend is also observed in the KDE curve of working hours for 7 kW, 24 kW, and 90 kW charging piles. In [Figure 3C](#), the Monte Carlo simulation results display energy indicators (DCR, RPR, V2B, and B2V) and carbon emission indicators (total carbon emissions from urban transportation and building electricity consumption). The numerical variances between DCR and RPR are minimal, typically differing by only three to four decimal places. The ranges for V2B and B2V are within −0.2% to 0.7% and −0.5% to 0.8%, respectively. The radar chart illustrates that the fluctuation range for carbon emissions in S5 is approximately ±0.1%.

[Figure 3D](#) presents the Monte Carlo simulation outcomes for the four economic indicators linked to [Figure 2](#). The results show that the

EVCS service profit, import cost, and export profit vary by only three to four decimal places. Additionally, the fluctuation range for electricity bill savings is approximately ±2%. [Figure 3](#) verifies the robustness of the urban building-E-mobility simulation platform using Nansha as an example, focusing on transportation, energy, environment, and economic indicators. The Monte Carlo simulation results demonstrate the platform's reliability in evaluating various aspects, such as transportation efficiency, RE penetration, environmental impact, and economic feasibility in the urban setting of Nansha. The rise in building electricity demand due to climate change and the power rationing policy under carbon neutrality goal may cause frequent power outages. This issue can be alleviated by implementing distributed RE systems, V2B/B2V energy interaction, and the climate change adaptation strategy for EVCS planning. [Figure 3E](#) illustrates the spatial distribution of building power outages in 2060 (left) and 2100 (right). In 2060, power outages for a single building range from 0 to 34 h, while in 2100, they range from 0 to 280 h.

[Figures 3F](#) and [3G](#) demonstrate the decrease in building power outage hours in 2060 and 2100 after the gradual implementation of distributed RE systems and V2B/B2V energy interaction, respectively. The maps and diagrams show that the RE can help decrease the power outage hours up to 10 h for an individual building in 2060 and 120 h in 2100. Besides, the maximum single-building power outage hours in 2060 decreases from 34 to 7 h by 79.4%, and in 2100, it decreased from 280 to 38 h by 86.4%. [Figure 3H](#) shows the reduction in power outage hours in 2060 and 2100 following the combination of distributed RE systems, V2B/B2V energy interaction, and the climate change adaptation strategy for EVCS planning. The maps and histograms show that compared with the reference strategy, the power outages are further reduced slightly in 2060 and greatly in 2100 with longer duration of a heat wave. Besides, compared with the reference strategy, the maximum single-building outage hours in 2060 decreased from 34 to 4 h by 88.2%, with a similar reduction in 2100. These results in [Figures 3E–3H](#) demonstrate that the climate change adaptation strategy can effectively reduce annual outage hours through V2B/B2V energy interaction and building energy management, even without employing a mandatory external signal to the EV fleet.

DISCUSSION

Electrified internet of energy enables multi-directional power interactions among building prosumers, E-mobility, and power grid for sustainable and smart city transformations. This study develops a city-scale information model for planning and distributions of urban infrastructures, together with an agent-based model urban energy system simulation platform for optimal allocations of EVCSs. Heavy computational loads through transient performance simulation have been well addressed via super-computing in high accuracy to assist the decision-making in city-scale sustainability transformations. Climate-adaptive resilience involves enhancing a system's capability to absorb, recover from, and adapt to the impacts of climate change.³⁷ This includes various actions across policy, infrastructure, services, planning, education, and communication. In this study, climate-adaptive resilience refers to the ability of EVs to supply excess electricity to buildings during high-temperature power outages, thereby reducing the duration of these outages. In Nansha, continuous high-temperature power outages are frequent during the summer, and the annual average number of power outage hours

(C) The distribution of EV battery capacity degradation percentage. For each scenario, the upper curve represents the probability density function (PDF) obtained by kernel density estimation (KDE) method, while the boxplot at the bottom shows the minimum, lower quartiles, median, upper quartiles, and the maximum value of EV battery capacity degradation percentage.

(D) The distribution of average daily working hours of 7-kW, 24-kW, and 90-kW charging piles in each scenario. The main body of the plot resembles two mirrored KDEs along the numerical axis. The shape and width of the violin body directly reflect the probability density at different values. The boxplot in a violin body functions the same as that in (C).

(E) Demand coverage ratio (DCR) and renewable penetration ratio (RPR) of Nansha in each scenario. The electricity demand for building operation and EV driving are considered. Renewable energy is generated by distributed building renewable energy systems, while the shortage is satisfied by coal-fired power grid. The value next to the red arrow indicates the percentage increase from the tail to the head in the direction of the arrow, while the value next to the cyan arrow indicates the percentage decrease from the tail to the head in the direction of the arrow.

(F) Sankey diagram of urban energy flow using S5 as an example.

(G) The left primary vertical axis is EVCS service profit of all scenarios in 10¹⁰ USD. The right vertical axis is the capital and maintenance cost of EVCSs of all scenarios in 10⁷ USD. The red arrows and cyan arrows hold the same meaning as those in (E).

(H) Carbon emissions for all scenarios are given in order of magnitude in a counterclockwise direction starting from S1. Scenarios where the charging station planning adopts a non-reference strategy are highlighted in light blue (ie, S5, S6, S7, S9, S11, S13).

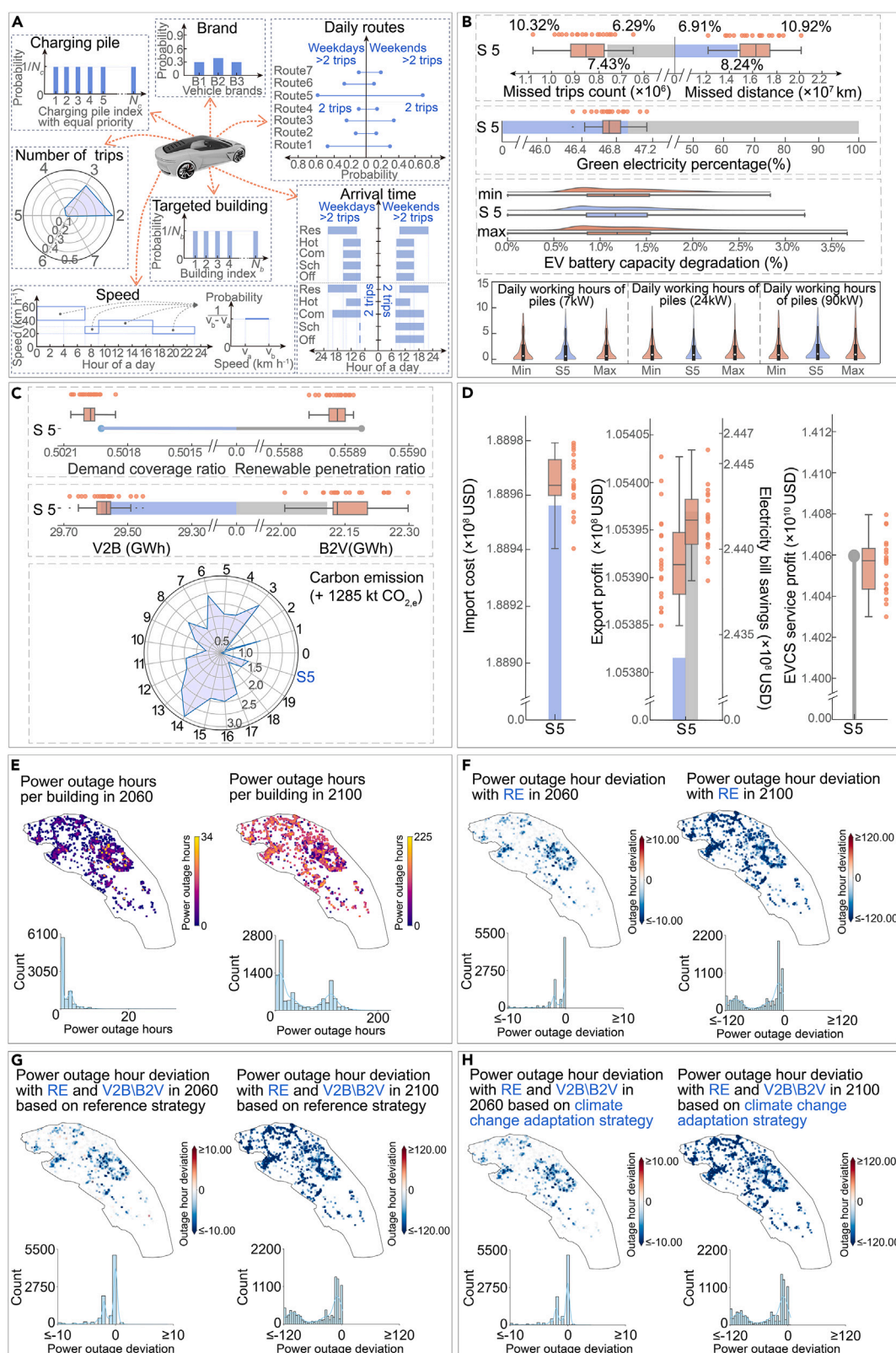


Figure 3. Uncertain variable description and Monte Carlo-based robustness evaluation

The charts that are named S5 and in light blue or gray color represent the results of S5 for comparison with other scenarios, while charts in orange color represent the Monte Carlo simulation results. The scatterplots show the indicator values of samples, while boxplots display the distribution of the data. The “min” and “max” refer to the minimum and maximum among results of samples.

(A) Uncertain variables and their probability distributions or intervals. Uncertain variables include the charging pile selection (ie, to determine one pile within piles with the same priority, and N_p denotes the number of piles with the same priority), the brand of an EV (ie, AION X, AION Y, and Tesla model Y), daily routes (see [Tables S6–S8](#)

(legend continued on next page)

serves as a quantitative measure for assessing climate change resilience. This study aims to enhance vehicle-to-building energy flow by optimizing the layout of charging stations, enabling EV fleets to supply energy when buildings face electricity shortages. Considering future climate change and extreme events, climate-adaptive resilience with energy supply reliability and performance robustness with parameter uncertainties have been evaluated. Specifically, the platform encompasses a building network (comprising five types of buildings equipped with storage batteries), a distributed RE system network (incorporating rooftop PV, building-integrated PV, and wind turbines), E-mobility fleets (consisting of three brands using two types of lithium batteries and accounting for operational capacity degradation), the charging infrastructure network, and the coal-fired power grid, facilitating energy interactions and communications between these components (also called agents). Through this platform, four EVCS strategies are formulated, addressing RE density, traffic flow, power outage hours, and building energy shortages. These strategies are evaluated and compared from perspectives of transportation performance (like missed trips and missed miles), RE penetration, carbon emissions, and economic feasibility. Furthermore, the platform's robustness is assessed through a Monte Carlo simulation, incorporating uncertainty models of EV behaviors within the context of the RE density strategy scenario. The study also examines the effects of climate change on building cooling, heating, and electricity demand (see [Figure S5](#)). Additionally, a climate change adaptation strategy aimed at reducing building power outage hours is validated in comparison to a reference strategy. This research contributes to the advancement of sustainable urban energy systems and offers valuable insights for policymakers and urban planners in the development of EV charging infrastructure and climate change adaptation strategies.

Key findings are summarized based on the simulation results. Electrified internet of energy with multi-directional energy interactions (V2B and B2V) can effectively enhance decarbonization and infrastructure efficiency in road traffic systems, while meeting transportation demand with minimum impact on EV battery capacity degradation. By developing the city-scale information model under the ABM framework, the proposed EVCS strategies outperform the reference strategy, facilitating increased penetration of RE in cities and buildings without external signals. Economic feasibility of the proposed EVCS strategies is demonstrated, with RE density and traffic flow strategies yielding the highest profits despite slightly higher costs. All strategies can effectively reduce import expenses, generate export profits, and provide significant electricity bill savings. Strategic EVCS planning supports decarbonization in urban building and transportation systems, with emissions varying across strategies, where S5 exhibits the lowest emissions. The climate change adaptation strategy effectively reduces emissions during power outages. The robustness of the urban building-E-mobility simulation platform is demonstrated in Nansha, evaluating transportation efficiency, RE penetration, environmental impact, and economic feasibility. Climate change drives increased cooling and electricity demand in Nansha, while the heating load remains consistent. The adaptation strategy reduces annual outage hours through V2B/B2V energy interaction and building energy management, without requiring a dispatch signal to the EV fleet.

Compared with previous studies, this paper advances in three important areas. First, in contrast to current studies focusing on the individual transportation sector, the proposed cross-sector building-transporta-

tion-charging infrastructure network energy simulation platform considers city-wide energy conservation and carbon reduction benefits resulting from inter-sector energy interactions. This offers insights for the development of future smart, green, and low-carbon cities. Second, unlike previous research limited to a few buildings, the ABM-based simulation platform examines energy interactions between buildings and electric transportation on a city scale, incorporating thousands of buildings and tens of thousands of EVs. This has practical implications for scheme implementation. Third, the platform's modeling of EV travel using probability and interval approaches, as opposed to deterministic models, enhances realism and energy performance robustness. The utilization of Monte Carlo simulation further strengthens the reliability of the conclusions. Further, the consideration of climate change's impact on urban energy systems distinguishes this work from common practices with typical year weather. This inclusion reflects the potential performance of charging infrastructure in future scenarios and underscores its role in enhancing urban energy climate resilience.

In this research, Nansha serves to validate the effectiveness of the proposed model and develop future charging station planning strategies aiming at sustainable city transformation. With adequate computational resources, this model can simulate energy networks on smaller scales, such as towns, as well as on larger scales, encompassing metropolises, provinces, and even entire countries due to the scale flexibility of the ABM framework.

Three significant topics for further investigation have been identified based on the limitations of method in this study. First, the optimization of EV fleet scheduling and coordination management within the urban energy systems is crucial. The random V2B/B2V energy interactions in the current simulation underutilize the potential for urban energy efficiency and decarbonization through building-E-mobility energy interactions. Strategically scheduling fleets that require charging to buildings with surplus renewable energy (RE) or directing fleets with excess electricity to support buildings lacking RE, could enhance RE penetration and minimize power outages. Second, a dynamic pricing strategy along with energy interactions is essential. The current assumptions of high environmental motivation among EV owners overlook practical incentives for participation. Introducing a well-designed pricing strategy could offer economic advantages, thus enhancing owners' willingness to engage in energy interactions. Finally, it is important to consider the societal effects of charging infrastructure. The current focus on technical, environmental, and economic evaluations overlooks the impact on social equity, such as accessibility and fairness among different demographic groups. Incorporating social indicators into future research could ensure the charging infrastructure reflects broader humanistic and social welfare considerations.

The proposed urban internet of energy platform within the ABM framework presents several challenges in real-world applications. These challenges include the demand for real-time data collection and updates, which call for a robust cyber-physical system, as well as the requirement for efficient data processing dependent on edge computing. Furthermore, it is crucial to manage minute-level variations in urban energy characteristics, such as fluctuations at supercharger stations. These requirements impose significant challenges on the storage and processing capabilities for urban spatiotemporal data.

for types of destinations of all daily routes), arrival time, a targeted building (ie, to determine one specific building within one building type.), speed, and the number of trips.

(B) Monte Carlo analysis results of traffic-side indicators, including missed trips' count, missed distance, green electricity percentage in EV charging, EV battery capacity degradation, and average daily working hours of charging piles at different power levels.

(C) Monte Carlo analysis results of energy performance indicators and carbon emissions, including DCR and RPR in Nansha and V2B/B2V energy interactions. The S5 is highlighted in light blue in the radar chart.

(D) Monte Carlo analysis results of economic performance indicators, including import cost, export profit, electricity bill savings, and EVCS service profit.

(E) Power outage hours per building in 2060 and 2100.

(F) Power outage hour deviation per building with renewable energy systems (ie, rooftop PV, BIPV, and wind turbine) in 2060 and 2100, relative to results in E.

(G) Power outage hour deviation per building with renewable energy systems (ie, rooftop PV, BIPV, and wind turbine) and energy interactions (ie, V2B and B2V) based on reference EVCS strategy in 2060 and 2100, relative to results in E.

(H) Power outage hour deviation per building with renewable energy systems (ie, rooftop PV, BIPV, and wind turbine) and energy interactions (ie, V2B and B2V) based on climate change adaptation strategy in 2060 and 2100, relative to results in E.

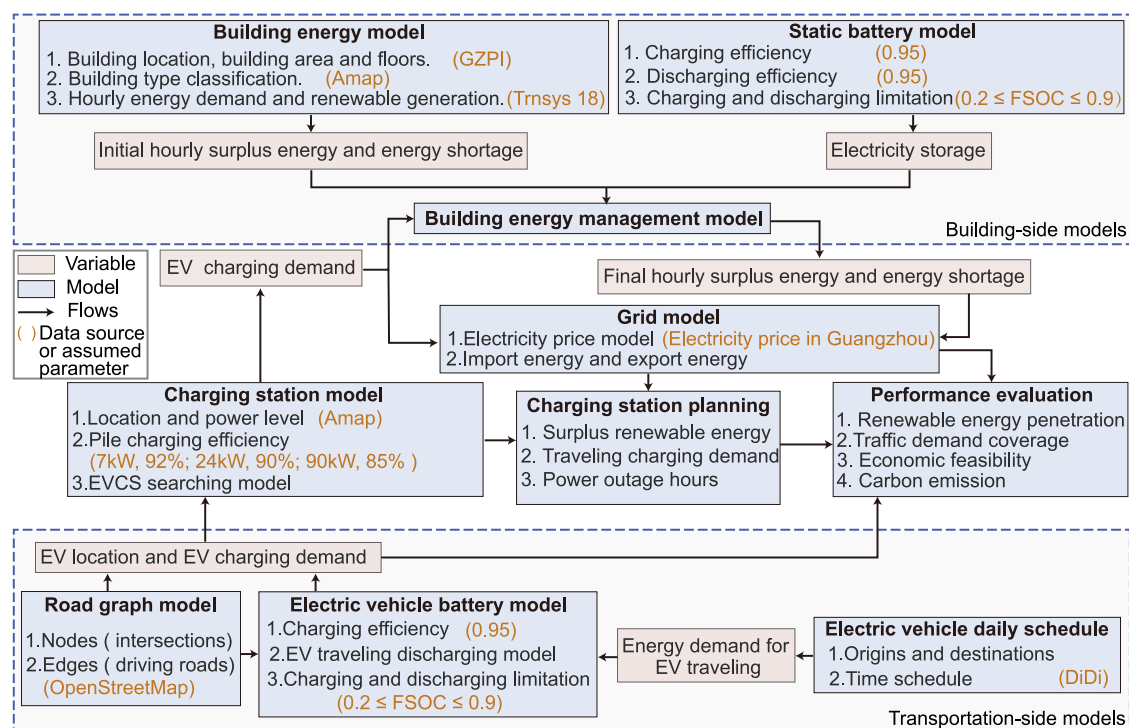


Figure 4. ABM simulation framework and models

METHODS

Overview of the city-scale information model under the ABM framework

Figure 4 presents the main algorithm flow chart of the ABM urban building-transportation-charging infrastructure network simulation platform (the detailed version is in Figures S6–S8). Within the ABM framework, the agents include buildings, EVs, charging stations, and power grids. The sub-models are developed to focus on the energy characteristics of these components and their interactions.

On the building side, the models include building energy models, static energy storage models, and building energy management models. The building energy model is created using the professional software TRNSYS 18, which simulates the hourly energy demand of the Nansha building complex and the renewable generation from the distributed RE system, integrating Nansha building geographic information system (GIS) data. The building energy management module manages energy scheduling based on surplus energy, energy shortages, the status of the energy storage system, and EV charging demand within each simulation time step. On the transportation side, the model includes an EV daily travel probability model, an EV battery charging and discharging model, and a road network model. The daily travel probability model is derived from a report by DiDi. The road network model, sourced from OpenStreetMap, calculates the actual distance between starting and ending points and provides node location information.

The current timestep's EV location and charging demand, combined with the charging station model (see Table S12), determine the EV charging demand at charging stations connected to the building RE system, thereby integrating with building energy management. EVCS information is collected from Amap, with the EVCS searching model detailed in Figure S9. Finally, EVCS spatial and power planning is conducted based on the distribution of surplus RE, EV travel charging demand, and simulated power outage hours. These planning strategies undergo techno-economic-environmental evaluations to provide EVCS planning recommendations based on different performance preferences.

District building energy demand and generation modeling

The coordinate, roof area, and building height data for Nansha buildings were supplied by the Guangzhou Urban Planning & Design Survey Research Institute (GZPI). Building type information was acquired through the point of interest (POI) data from the Amap Open Platform. Studied buildings include 3,540 residential buildings, 2,800 office buildings, 1,215 commercial buildings, 1,095 school buildings, and 399 hotels.

The annual energy demand data for each building within each type is converted based on the ratio of area to that of the TRNSYS model (see Tables S13–S15 and Note S2), according to the following (Equation 1):

$$L_{i,t} = L_{\text{typical},t} / S_{\text{typical}} \times S_i \quad (\text{Equation 1})$$

where $L_{i,t}$ represents the energy demand of building i at timestep t , $L_{\text{typical},t} / S_{\text{typical}}$ represents energy demand per unit area, and S_i denotes some kind of area of building i . In the context of building energy demand, S_i represents the gross building area. The method for converting building RE generation follows the same principle as energy demand, using ratios from the TRNSYS model of each building type. For rooftop photovoltaic (PV) systems, S_i denotes the rooftop area of the building, as these systems are installed on roofs. In the case of building-integrated photovoltaics (BIPV), S_i refers to the total surface area of the building's four exterior walls, since BIPV systems are integrated into the facades. Each building is assigned a 100-kW and a 60-kW wind turbine (see Table S16, Figures S10 and S11, and Note S2).

EVCS search

When an EV requires charging, the strategy to find a charging station varies based on its location and fractional state of charge (FSOC). If the EV's FSOC drops below 0.2, it is considered to have failed and will be towed back to its residential building. For an EV in transit with an FSOC below 0.7 and more than 1 h remaining to reach the next destination, the search for a charging station is initiated. The decision on where to charge depends on the EV's proximity to its destination and its current FSOC: if the EV is close to the destination and the FSOC is 0.4 or

below, it will be charged nearby; if the FSOC is greater than 0.4, it will seek charging near the destination. Otherwise, it will charge at the nearest available station. For EVs stationary at their departure node that require charging, the search will focus on nearby charging stations. Detailed strategies for locating EVCSs, whether near a building or in route (see Figure S9), along with the flow chart of EVCS searching, are shown in Figure S7.

EV battery model

The ABM simulation framework integrates models that simulate the charging and discharging process of both building static batteries and EV power batteries. Static batteries receive their charge from RE sources and provide power to the building during periods when RE generation is inadequate. The models are elaborated in Equations 2 and 3 for charging and discharging, respectively:

$$FSOC_{i,t1} = FSOC_{i,t0} + \frac{eta_c \times P_{i,c} \times \text{timestep}}{Cap_{i,max} \times RC_i} \quad (\text{Equation 2})$$

$$FSOC_{i,t1} = FSOC_{i,t0} - \frac{eta_d \times P_{i,d} \times \text{timestep}}{eta_d \times Cap_{i,max} \times RC_i} \quad (\text{Equation 3})$$

where $FSOC_{i,t1}$ and $FSOC_{i,t0}$ denote the FSOC of the i th EV before and after the timestep respectively; eta_c and eta_d refer to the charging efficiency and discharging efficiency, respectively; and $P_{i,c}$ and $P_{i,d}$ indicate the charging power and discharging power, respectively. $Cap_{i,max}$ denotes the full capacity of the i th EV battery and RC_i represents the relative capacity percentage of the i th EV battery. The parameters and energy management of static batteries are given in Table S17.

For EV power batteries, the situations differ. These batteries can be charged using surplus RE generated by buildings or electricity sourced from coal-fired power grids at charging piles. Furthermore, at bidirectional charging piles, EV power batteries have the capability to discharge back to buildings when the RE is inadequate or discharging when driving on the road. The vehicle-to-building (V2B) discharge model for EV batteries mirrors that of static batteries, with the distinction being that the effective discharge power is determined by the lesser of the two values: the battery's discharge power or the power level of the bidirectional charging pile. The same pattern can be found in the charging model of EV batteries. The discharging model of driving is shown in Equations 4 and 5:

$$factor_{drive} = \frac{FSOC_{up} - FSOC_{low}}{D_{i,max}} \quad (\text{Equation 4})$$

$$FSOC_{i,t1} = FSOC_{i,t0} - factor_{drive} \times D_{drive} \quad (\text{Equation 5})$$

where $FSOC_{up} - FSOC_{low}$ denoted the available FSOC, $FSOC_{up}$ equals 0.9 while $FSOC_{low}$ equals 0.2; $D_{i,max}$ refers to the maximum cruising range of i th EV, and D_{drive} indicates the distance the EV has traveled this time. The detailed parameters and energy management of EV batteries can be found in Tables S3, S4, S8–S10, and S18.

There are two types of lithium batteries in this ABM simulation platform, involving $LiFePO_4$ and $LiNiCoAlO_2$ with different degradation curve. The capacity degradation model is developed based on linear interpolation for adjacent two DoD curves,⁴⁰ shown as follows:

$$DoD_{peak, valley} = FSOC_{peak, EFC} - FSOC_{valley, EFC} \quad (\text{Equation 6})$$

$$RC_t = \frac{Cap_t}{Cap_{initial}} \quad (\text{Equation 7})$$

$$RC_{DoD} = k_1 EFC^3 + k_2 EFC^2 + k_3 EFC + k_4 \quad (\text{Equation 8})$$

$$EFC_{t+t} = EFC_t + DEFC_{t+t} \quad (\text{Equation 9})$$

$$DEFC_{t+t} = \frac{DEnergy}{Cap_{initial} \times 2} = \frac{|FSOC_{t+t} - FSOC_t| \times Cap_t}{Cap_{initial} \times 2} \quad (\text{Equation 10})$$

$$RC_{DoD, t+\tau} = \frac{DoD_{t+\tau} - DoD_2}{DoD_1 - DoD_2} \times RC_{DoD1, t+\tau} + \frac{DoD_{t+\tau} - DoD_1}{DoD_2 - DoD_1} \times RC_{DoD2, t+\tau} \quad (\text{Equation 11})$$

where EFC means equivalent full cycles where a cycle refers to the accumulated charging or discharging energy equals to two times of initial capacity. Equation 11 denotes the linear interpolation between $(DoD_1, RC_{DoD1, t+\tau})$ and $(DoD_2, RC_{DoD2, t+\tau})$ to get $RC_{DoD, t+\tau}$. The coefficient parameters of $LiFePO_4$ and $LiNiCoAlO_2$ are given in Tables S3 and S4, respectively.

Road traffic network modeling

The road traffic network model was extracted from OpenStreetMap and simplified into an undirected graph model, allowing for bidirectional reachability between any two nodes connected by edges. The simplified Nansha road traffic network comprises 5,830 nodes and 8,297 edges, where nodes symbolize road intersections and serve to indicate the positions of buildings and charging stations within the network. The edges of this network represent the actual road distance between the two linked nodes, rather than the straight-line distance. To determine the shortest path between a departure node and destination node, the Floyd-Warshall algorithm is utilized to compute the optimal route within the road network.

Energy interaction strategies

This study comprises five energy management systems (EMSs): (1) EMS for buildings without static batteries; (2) EMS for buildings with static batteries but no interactions with EVs; (3) EMS for buildings with both static batteries and EV interactions; (4) EMS for EVCS during EV charging from buildings with the most surplus RE during B2V process, or G2V; and (5) EMS for EVCS during V2B processes where EVs supply electricity to buildings with the largest RE shortage. Detailed rule priority diagrams and explanations can be found in Figures S12 and S13.

Evaluation criteria in technology, economy, and environment

$$DCR = 1 - \frac{\int_0^{t_{end}} P_{imp, bd}(t) dt}{\int_0^{t_{end}} D_{bd}(t) dt} \quad (\text{Equation 12})$$

Demand coverage ratio (DCR) is shown in Equation 12, where $D_{bd}(t)$ denotes the energy demand of a building, and DCR means the ratio of the energy demand covered by renewable generation to the total energy demand; $P_{imp, bd}(t)$ denotes the power imported from the grid by buildings.

$$RPR = 1 - \frac{\int_0^{t_{end}} P_{exp, bd}(t) dt}{\int_0^{t_{end}} P_{RE}(t) dt} \quad (\text{Equation 13})$$

Renewable penetration ratio (RPR) is shown in Equation 13, where $P_{exp, bd}(t)$ denotes the surplus renewable generation power exported to the grid by buildings; $P_{RE}(t)$ denotes the renewable generation power, and RPR means the ratio of the renewable generation consumed by buildings to the total renewable generation.

$$Trips_{miss} = \sum_{EV=0}^{EV_{tot}} trips_{missed}(EV) \quad (\text{Equation 14})$$

The number of missed trips is shown in Equation 14, where $trips_{missed}(EV)$ denotes the number of missed trips of an EV within 1 year, and $Trips_{miss}$ denotes the total number of missed trips of all EVs within 1 year.

$$Miles_{tot, miss} = \sum_{EV=0}^{EV_{tot}} Miles_{missed}(EV) \quad (\text{Equation 15})$$

The formula of missed miles is shown in Equation 15, where $Miles_{missed}(EV)$ denotes the missed miles of one EV. $Miles_{tot,miss}$ denotes the total number of missed miles of all EVs within 1 year.

$$T_{cp,daily} = \frac{\int_0^{t_{end}} T_{cp,work}(t) dt}{day_{tot}} \quad (\text{Equation 16})$$

The average number of hours a charging pile is used per day is shown in Equation 16, where $T_{cp,work}(t)$ denotes the charging pile working time-step (ie, working for 1 h), day_{tot} denotes the total number of days, and $T_{cp,daily}$ denotes the average number of hours a charging pile is used per day.

$$C_{imp,bd} = \int_0^{t_{end}} [P_{imp,bd}(t) \times Price_{imp,bd}(t)] dt \quad (\text{Equation 17})$$

Building import cost (for one building of one building type in a year) is shown in Equation 17, where $C_{imp,bd}$ denotes import cost of buildings. $P_{imp,bd}(t)$ denotes grid import power of buildings. $Price_{imp,bd}(t)$ denotes the electricity price in Guangzhou (different for residential and school buildings and the rest of the types of buildings; see Tables S19 and S20).

$$C_{imp,cp} = \int_0^{t_{end}} [P_{imp,cp}(t) \times Price_{imp,cp}(t)] dt \quad (\text{Equation 18})$$

EVCS import cost is shown in Equation 18, where $C_{imp,cp}$ denotes import cost of charging piles. $P_{imp,cp}(t)$ denotes grid import power of charging piles. $Price_{imp,cp}(t)$ denotes the commercial electricity price for public charging piles in Guangzhou (see Table S20).

$$C_{ini,cap} = \sum_1^{max(x)} Cap_x \times Price_{capital,x} \quad (\text{Equation 19})$$

$$C_{ini,num} = \sum_1^{max(cp)} Num_{cp} \times Price_{capital,cp} \quad (\text{Equation 20})$$

Total cost is calculated by Equations 18, 19, 20, 21, 22, and 23, where $C_{ini,cap}$ denotes initial capital cost of rooftop PVs, BIPVs, WTs, and batteries; Cap_x denotes the capacity of x , and the x represents rooftop PVs, BIPVs, WTs, or batteries. $Price_{capital,x}$ denotes the unit capital price of x (see Table S21). Num_{cp} denotes the number of charging piles, and $Price_{capital,cp}$ denotes the unit price of a charging pile. $C_{ini,num}$ denotes the total initial cost of charging piles.

$$C_{O\&M,cap} = \sum_1^{max(x)} Cap_x \times Price_{O\&M,x} \quad (\text{Equation 21})$$

$$C_{O\&M,num} = \sum_1^{max(cp)} Num_{cp} \times Price_{O\&M,cp} \quad (\text{Equation 22})$$

where $C_{O\&M,cap}$ denotes maintenance cost of rooftop PVs, BIPVs, WTs, and batteries and $Price_{O\&M,x}$ denotes the unit maintenance price of x (see Table S21). $Price_{O\&M,cp}$ denotes the unit price of a charging pile. $C_{O\&M,num}$ denotes the total maintenance cost of charging piles.

$$C_{tot} = C_{ini,cap} + C_{ini,num} + C_{O\&M,cap} + C_{O\&M,num} \quad (\text{Equation 23})$$

where C_{tot} denotes the total cost of the system composed of distributed renewable generation equipment, static lithium batteries, and charging infrastructure.

The service profit of a charging pile is calculated as follows:

$$Ser_{cp} = \sum_1^{max(cp)} Price_{ser} \times E_{cp} \quad (\text{Equation 24})$$

$Price_{ser}$ denotes the service price of charging piles, which is assumed to be 0.11 USD $\cdot$ kWh⁻¹ according to the actual service price in Nansha. E_{cp} denotes the total charging energy of the charging pile. Ser_{cp} denotes the total potential profit in the EVCS service market.

The equation of electricity bill savings of a building, $Save_{bd}$ is as below:

$$Save_{bd} = \int_0^{t_{end}} [P_{RE,bd}(t) \times Price_{imp,bd}(t)] dt \quad (\text{Equation 25})$$

$P_{RE,bd}(t)$ denotes the power of RE generation that is used for building energy demand coverage. $Price_{imp,bd}(t)$ denotes the price of electricity in Guangzhou.

The equation of export profit of a building, $Profit_{exp,bd}$ is as below:

$$Profit_{exp,bd} = \int_0^{t_{end}} [P_{exp,bd}(t) \times Price_{exp,bd}(t)] dt \quad (\text{Equation 26})$$

$P_{exp,bd}(t)$ is the power of RE generation that is exported to the grid. $Price_{exp,bd}(t)$ denotes the price of exporting, which is assumed to be 0.065 USD $\cdot$ kWh⁻¹ (ie, benchmark price for coal in Guangzhou).⁴¹

$$CE_{bd} = \int_0^{t_{end}} [P_{imp}(t) \times CEF] dt \quad (\text{Equation 27})$$

Operational carbon emission of a building is calculated by Equation 27, where CE_{bd} denotes operational net equivalent carbon emission (kg CO_{2,e} kWh⁻¹) of one building of one building type in a year. $P_{imp}(t)$ denotes grid import power of the building. CEF denoted average carbon emission factor of the grid in Guangzhou,⁴² 0.6379 kg CO_{2,e} kWh⁻¹.

$$CE_{cp} = \int_0^{t_{end}} [P_{imp,cp}(t) \times CEF] dt \quad (\text{Equation 28})$$

Operational carbon emission of a charging pile is calculated by Equation 28, where CE_{cp} denotes net equivalent carbon emission (kg CO_{2,e} kWh⁻¹) of a charging pile. $P_{imp,cp}(t)$ denotes grid import power of a charging pile. CEF denotes the average carbon emission factor of the grid in Guangzhou, 0.6379 kg CO_{2,e} kWh⁻¹. Then total carbon emission of building and transportation sector in Nansha is calculated by

$$CE_{total} = \int_0^{max(bd)} CE_{bd} + \int_0^{max(cp)} CE_{cp} \quad (\text{Equation 29})$$

Where CE_{total} denotes net equivalent carbon emission (kg CO_{2,e} kWh⁻¹). $max(bd)$ denotes the maximum index of buildings. $max(cp)$ denotes the maximum index of charging piles.

RESOURCES AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Dr. Yuekuan Zhou (yuekuan.zhou@outlook.com; yuekuanzhou@hkust-gz.edu.cn).

Materials availability

This study did not generate new unique materials.

Data and code availability

The authors declare that the data used as model inputs supporting the findings of this study are available within the paper and its supplemental information files. In addition, all data and calculation tables have been uploaded to <https://zenodo.org/records/13325018>. This study did not generate new unique code.

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AUTHOR CONTRIBUTIONS

Conceptualization, Z.D. and Y.Zhou; methodology, Z.D., Y.Zhou, and A.S.; software, Z.D. and A.S.; validation, Z.D. and A.S.; writing—original draft, Z.D., A.S., Y.Zheng, and X.Z.; writing – review & editing, Y.Zhou; funding acquisition, Y.Zhou; supervision, Y.Zhou.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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YNEXUS, Volume 2

Supplemental information

City information models for optimal EV charging and energy-resilient renaissance

Zhaohui Dan, Aoye Song, Yuyu Zheng, Xinyue Zhang, and Yuekuan Zhou

O: Office R: Residential C: Commerical H: Hotel S: School

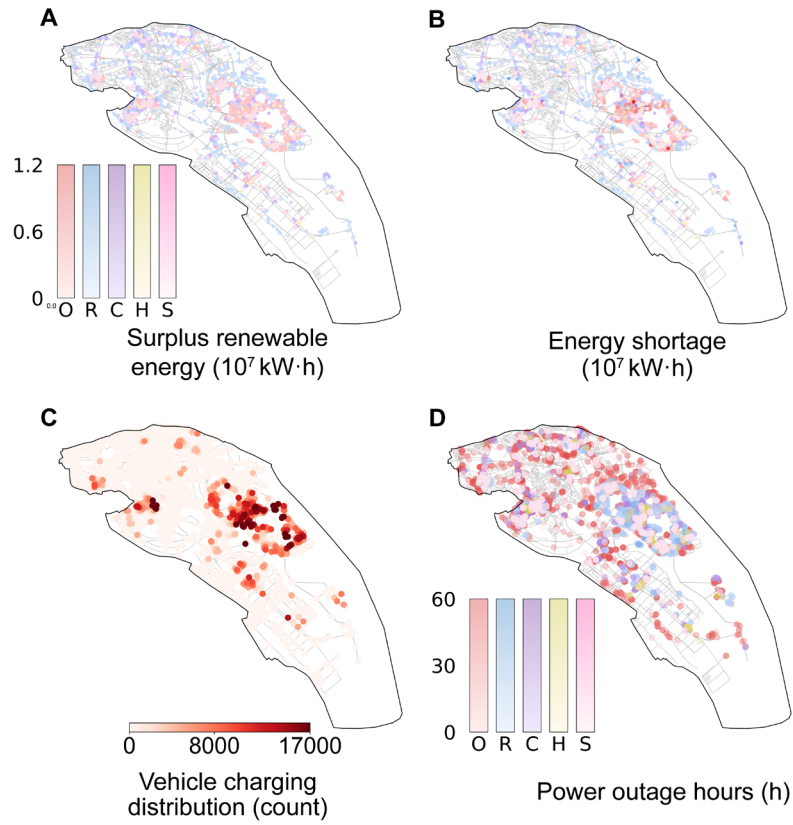


Figure S1 Maps of surplus energy of buildings, energy shortage of buildings, vehicle charging counts, and power outage hours.

(A) Maps of surplus renewable energy.

(B) Energy shortages.

(C) Distribution of vehicle charging counts.

(D) Power outage hours.

Figure S1 shows the data basis for the four proposed EVCS planning strategies.

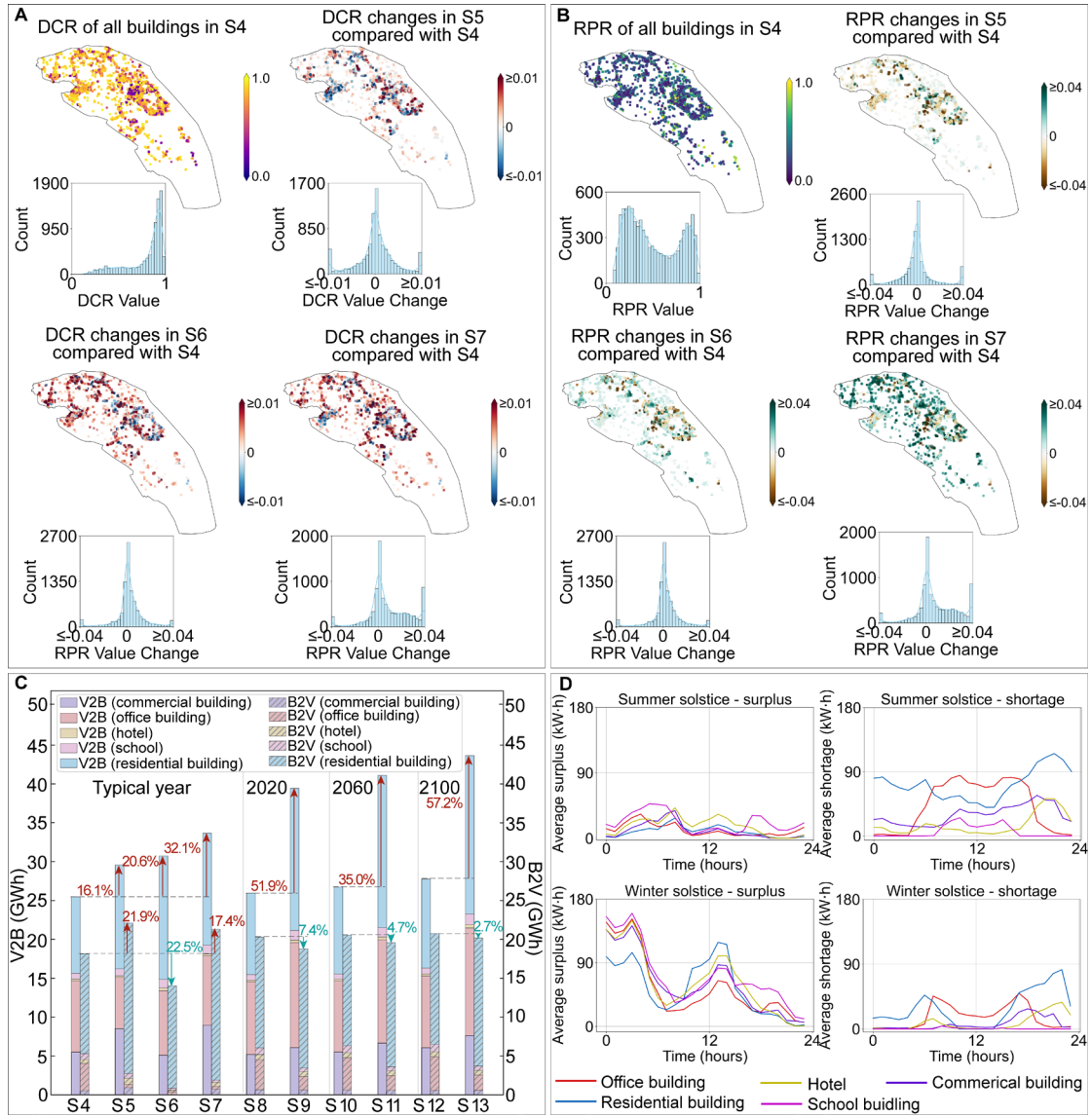


Figure S2 Energy performance indicators for all scenarios.

(A), Four heatmaps along with frequency histograms show DCR of each building in S4, and the changes compared to S4 in S5, S6, and S7, respectively.

(B), Four heatmaps along with frequency histograms show RPR of each building in S4, and the changes compared to S4 in S5, S6, and S7, respectively.

(C), The total energy of V2B and B2V energy interactions of each scenario are shown in stack bars, consisting of the energy from or to commercial buildings, office buildings, hotels, school buildings, and residential buildings, respectively. The red upward arrow represents the percentage increase based on the value of the dotted line, and the cyan downward arrow represents the percentage decrease.

(D), Time series curves of average surplus energy and energy shortage of five types of buildings at summer and winter solstice with hourly resolution.

At the building scale, in **Figure S2A**, as shown in the maps and histograms, compared to the DCR in S4, the S5, S6, and S7 show improvement in DCR where S7 achieves the highest improvement magnitude. This is because the mix strategy in S7 considers both the surplus energy of buildings and charging behaviours of EVs to maximize the V2B interactions (as shown in **Figure S2C**), and therefore the energy directly imported from the grid decreases, leading to the improvement of DCRs. Same trend can be found in RPR at the building scale in **Figure S2B**. **Figure S2D** shows that regardless of the summer or winter solstice, office buildings have the largest energy shortage during the day, while residential buildings have the largest energy shortage at night. In addition, residential buildings and hotels have more surplus energy during the day, while office buildings and school buildings have more surplus energy at night. This is because of the functional characteristics of different types of buildings.

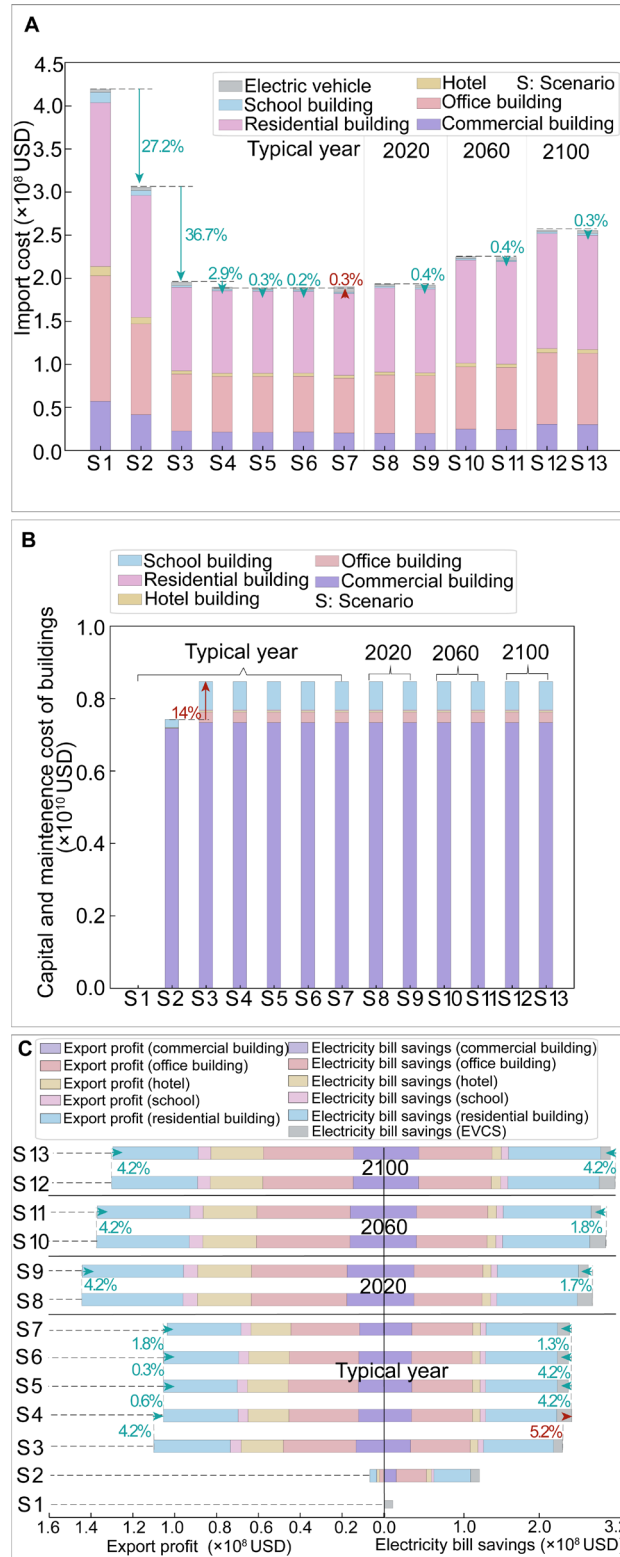


Figure S3 Economic performance indicators for all scenarios.

(A) Import cost of scenarios in 10^8 USD.

(B) The vertical axis is the capital and maintenance cost of buildings of all scenarios in 10^{10} USD.

(C) The left primary vertical axis is the profit by exporting electricity to the grid of all scenarios in 10^8 USD. The right vertical axis is the electricity bill savings of all scenarios compared to fully relying on the coal-fired power grid in 10^8 USD.

In **Figure S3A**, the distributed PV and wind turbine systems, energy interactions, and EVCS planning measures can help reduce grid import costs. Specifically, S1 to S3 show step-by-step

cost reductions of 27.2% and 36.7% due to the installation of renewable energy equipment. Further reduction can be seen in S5 (renewable density strategy), and S6 (traffic flow strategy) of \$570,000 and \$380,000 due to energy interactions and EVCS planning. Similar trends can also be found in S9 (2020), S11 (2060), and S13 (2100), with the import costs dropped by \$768,000, \$900,000, and \$753,000, respectively. The cost reduction potential of proposed EVCS strategies can be further enhanced with EV fleet scheduling (future research). In **Figure S3B**, future Nansha's distributed renewable energy system costs around \$9 billion for capital investment and maintenance in the first year. Additionally, in the typical year, compared to S4 (reference strategy), the EVCS cost in S6 (traffic flow strategy) is 29% lower, while the EVCS costs in S5 (renewable density strategy) and S7 (mix strategy) are 8% and 15% higher. This advantage of S6 is due to S6 prioritizing EV charging over energy interactions, resulting in fewer charging piles than S4 with limited space demand. In contrast, S5 emphasizes RE density and interactions, both of which are related to widespread buildings, leading to slightly more charging piles than S4. In the climate change adaptation strategy, EVCS costs decrease by 11%, with fewer charging piles addressing power outages and energy shortages. **Figure S3C** shows that the proposed EVCS strategies yield lower export profits than the reference strategy. This is because that the amount of grid exporting RE decreased due to the improved RE utilization efficiency via V2B and B2V interactions. Besides, electricity bill savings drop by 1.3% to 4.2% compared to the reference, attributed to reduced B2V and increased V2B interactions (**Figure 2C**). This signifies that EVs tend to charge more from the grid to aid RE-deficient buildings.

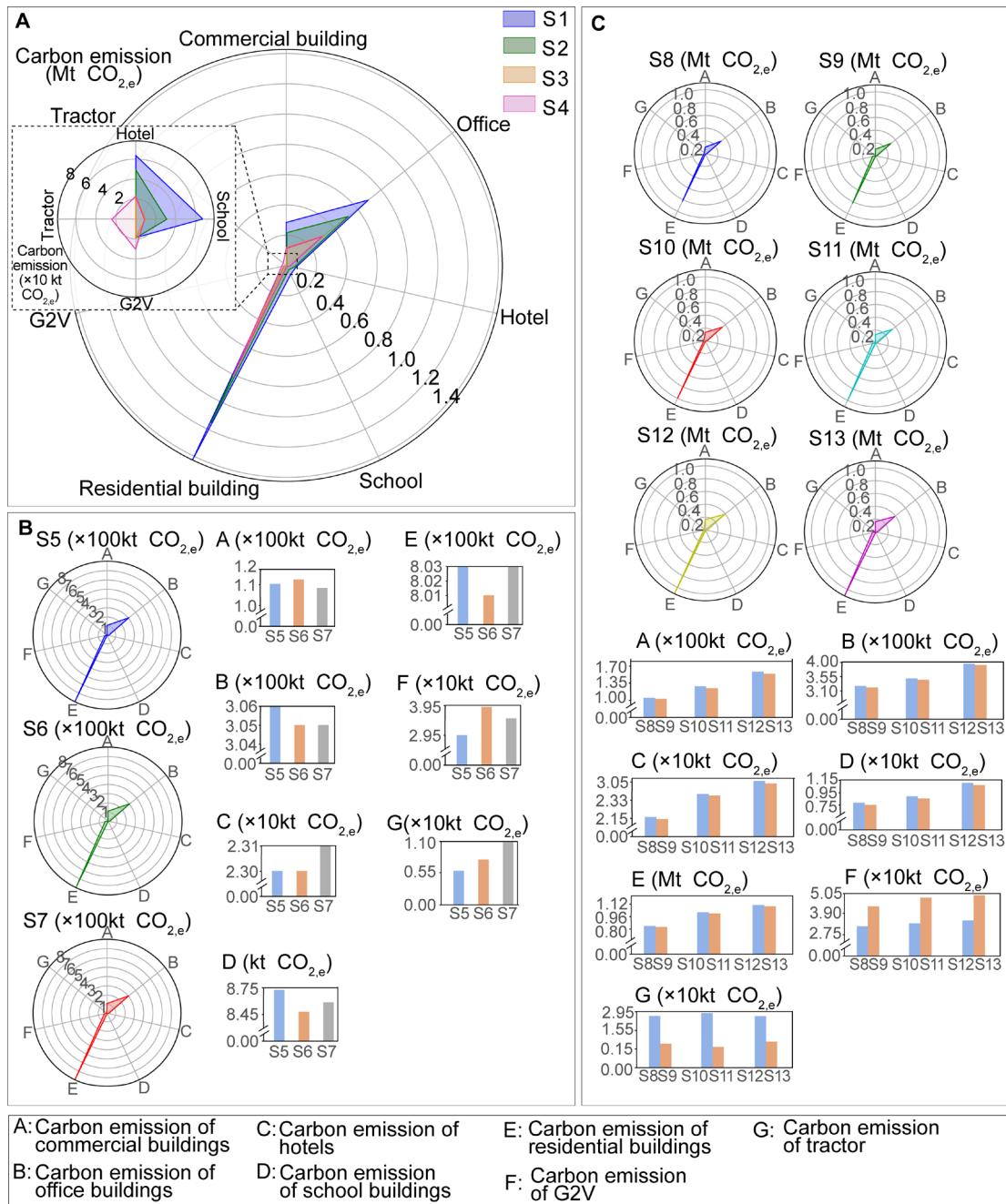


Figure S4 Environmental performance indicators for all scenarios.

(A) The big radar chart shows the components of carbon emissions of S1, S2, S3, and S4 in terms of commercial building, office building, hotel, school building, residential building, EV charging from the grid, and the tractor in units of Mt CO_{2,e}. The small radar chart is an amplification of the carbon emissions of school building, hotel, EV charging from the grid and tractor in units of 10kt CO_{2,e}.

(B) The radar charts show the components of carbon emissions of S5, S6, and S7 in terms of commercial building, office building, hotel, school building, residential building, EV charging from the grid, and the tractor. The bar charts show the carbon emissions of each component of the three scenarios in different units.

(C) The radar charts show the components of carbon emissions of S8-S13 in terms of commercial building, office building, hotel, school building, residential building, EV charging from the grid, and the tractor. The bar charts show the carbon emissions of each component of the six scenarios in different units.

In **Figure S4A**, carbon emissions are the highest from residential, office, and commercial buildings due to building numbers, energy demand, and limited RE sources. Emissions from hotels and schools gradually decrease by 10 kt CO_{2,e} in S1-S4, while G2V emissions remain

consistent in S1-S3. The capacity increases in S1-S3 and energy interactions in S4 boost the RE ratio in building power supply. In S4, EVs charge from the coal-fired grid at lower prices to support energy-deficient buildings. Besides, tractor emissions (carbon emissions generated by towing EVs back to residential buildings when EV's FSOC is less than 0.2) during transit increase due to the distance of increased missed trips (**Figure 2A**). In **Figure S4B**, the radar charts reveal that in the typical year, carbon emissions exhibit a comparable distribution across the 7 aspects as observed in S1-S4 (large radar chart in **Figure S4A**). Furthermore, the bar charts demonstrate that in S5-S6, the carbon emissions for each aspect display varying ranges within the same unit. This suggests that S5-S7 exhibit similar performance in carbon emissions, both in terms of distribution and quantity. The results in **Figure S4C** highlight the significant impact of climate change on carbon emissions and the effectiveness of the climate change adaptation strategy. Bar charts A-E demonstrate a gradual increase in emissions in 2020, 2060, and 2100, with S9, S11, and S13 consistently lower than the reference strategy. In contrast, bar chart F shows higher emissions from the adaptation strategy due to the support for energy-deficient buildings and reduced power outage hours through V2B. The G chart relates to missed trips in **Figure 2A** and distances travelled by tractors for EVs.

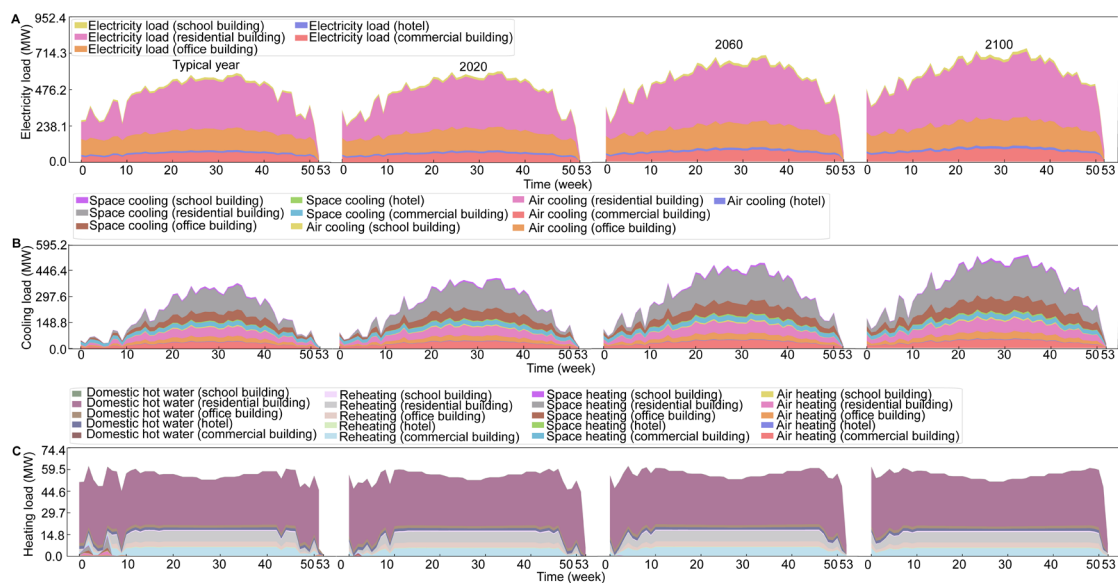


Figure S5 Cooling/heating/electric demands in city-level district communities.

(A) Stacked plots of the electricity load of five types of buildings (i.e., school building, residential building, office building, hotel, and commercial building) at weekly resolution in the typical year, 2020, 2060, and 2100.

(B) Stacked plots of the space cooling load and air-cooling load of five types of buildings (i.e., school building, residential building, office building, hotel, and commercial building) at weekly resolution in the typical year, 2020, 2060, and 2100.

(C) Stacked plots of the heating load of five types of buildings (i.e., school building, residential building, office building, hotel, and commercial building) at weekly resolution in the typical year, 2020, 2060, and 2100. The heating load consists of domestic hot water heating load, reheating load, space heating load, and air heating load.

Cooling/heating/electric demands in city-level district communities are highly dependent on climate change. Changes in temperature, humidity, wind speed and other variables can lead to the increase in demand for cooling, heating and electricity in buildings year by year. In **Figure S5A**, the time series loads of five different types of buildings are depicted, showcasing an increase from a typical year to 2020, 2060, and 2100 because of climate change, in which residential buildings, office buildings, and commercial buildings contribute to the top 3. The building electricity load is influenced by electric cooling, electric heating, and other electrical equipment. As temperatures rise, the workload on the building's cooling/heating system increases, leading to a higher demand for electricity. In **Figure S5B**, a similar growth trend is observed in the space cooling and air-cooling demand of the five types of buildings. Notably, the space cooling demand of residential buildings accounts for the highest proportion. This trend highlights the increased cooling demands in buildings, particularly in residential structures, as temperatures rise due to climate change. In **Figure S5C**, compared to cooling load and electricity load, heating load is less obviously affected by climate change. Because Nansha is in the subtropical region with hot summers and warm winters, the demand for space heating in winter is relatively low. The stacking chart reveals that for all buildings, domestic hot water and reheating loads represent a larger proportion than space heating and air heating. Specifically, domestic hot water load, reheating load of residential buildings, and reheating load of commercial buildings are identified as the top three largest sources of heating load. **Figure S5** suggests a growing demand for cooling and electricity in buildings in Nansha because of climate change, while the heating load remains relatively consistent.

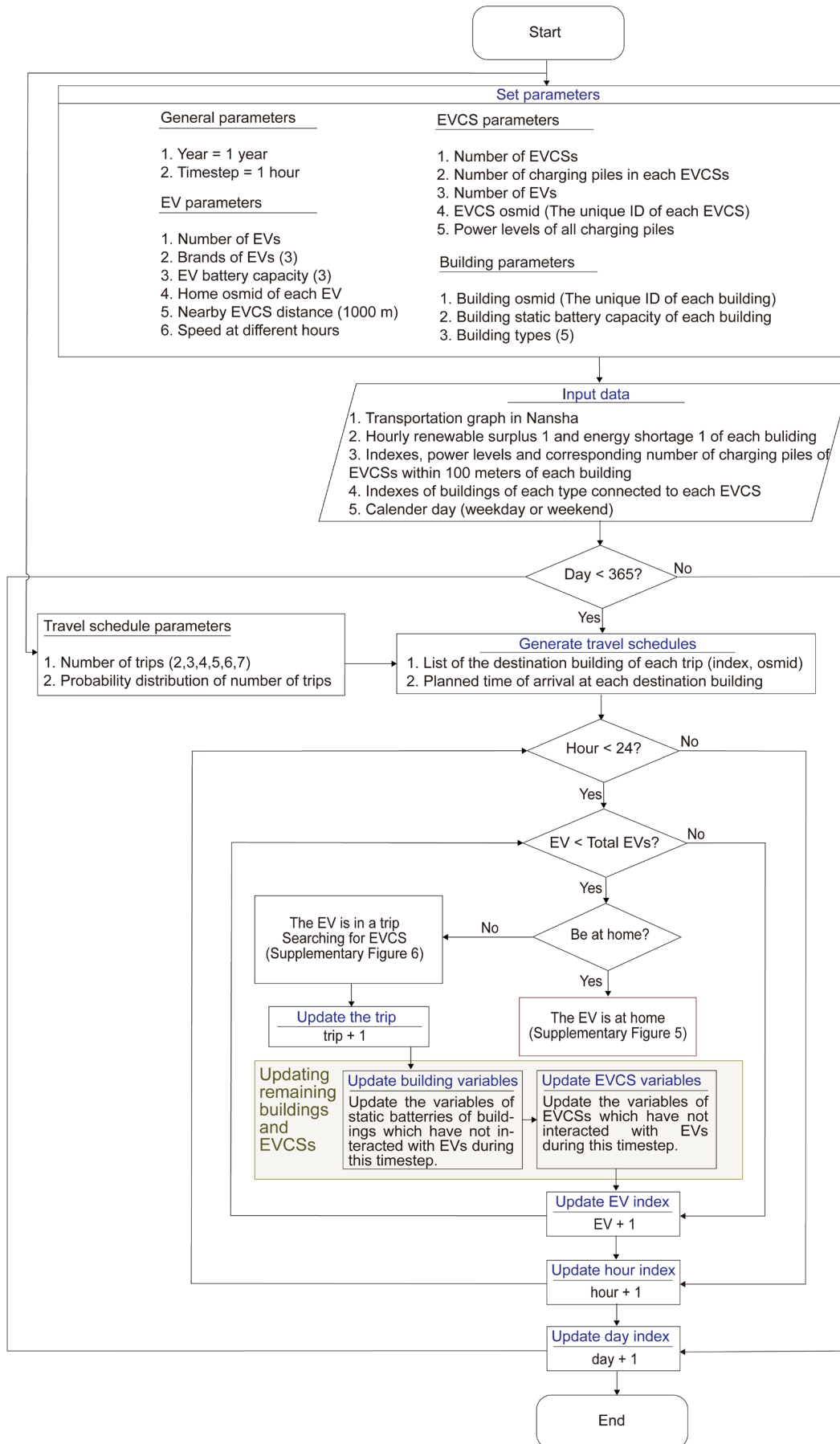


Figure S6 The methodology flow chart. The research process of this study is divided into four steps, including setting parameters, input data, simulation of building-transportation-grid

cross-energy system for one year, and evaluation of techno-economic-environmental performances.

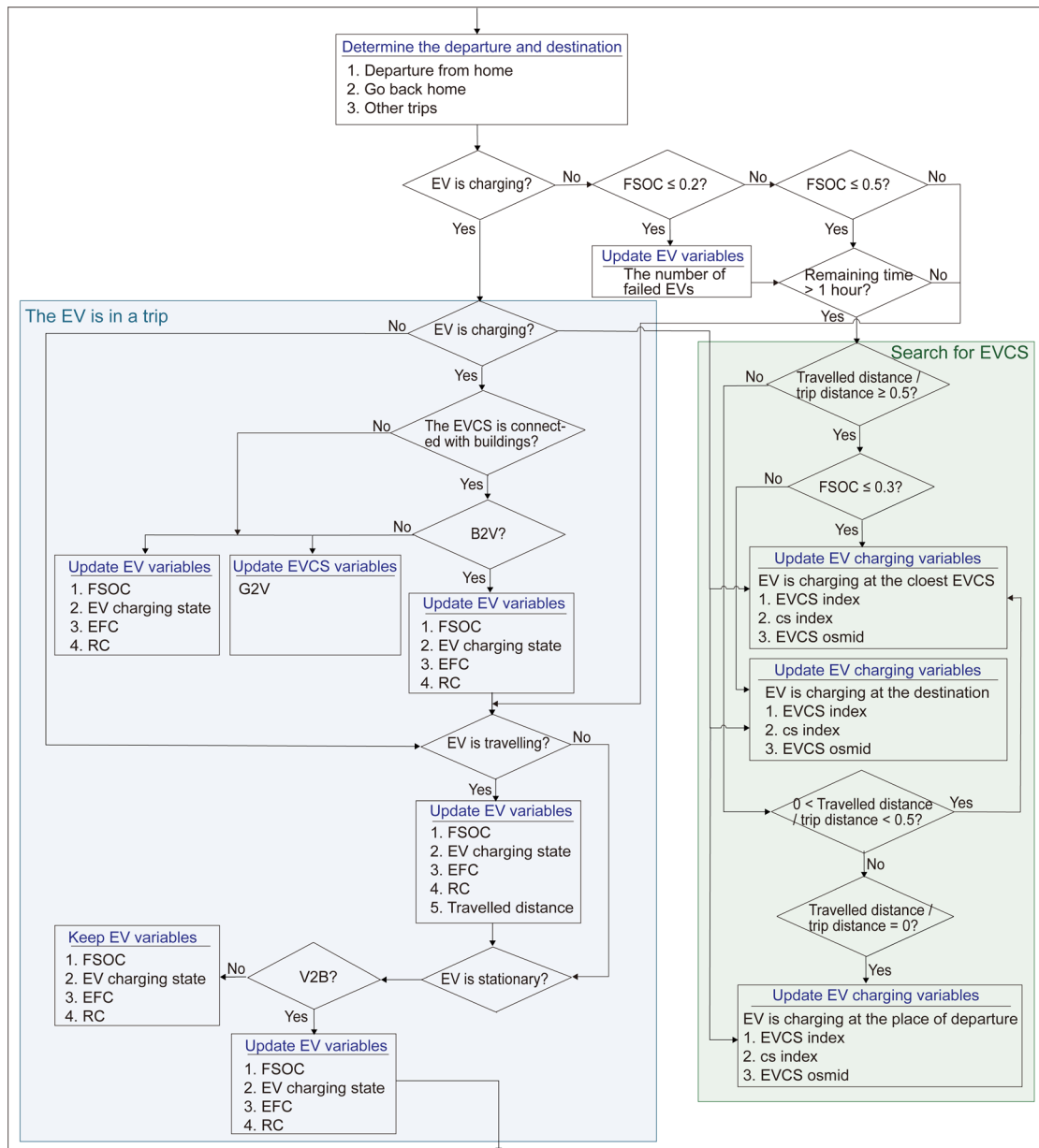


Figure S7 Methodology flow chart (EV at trip).

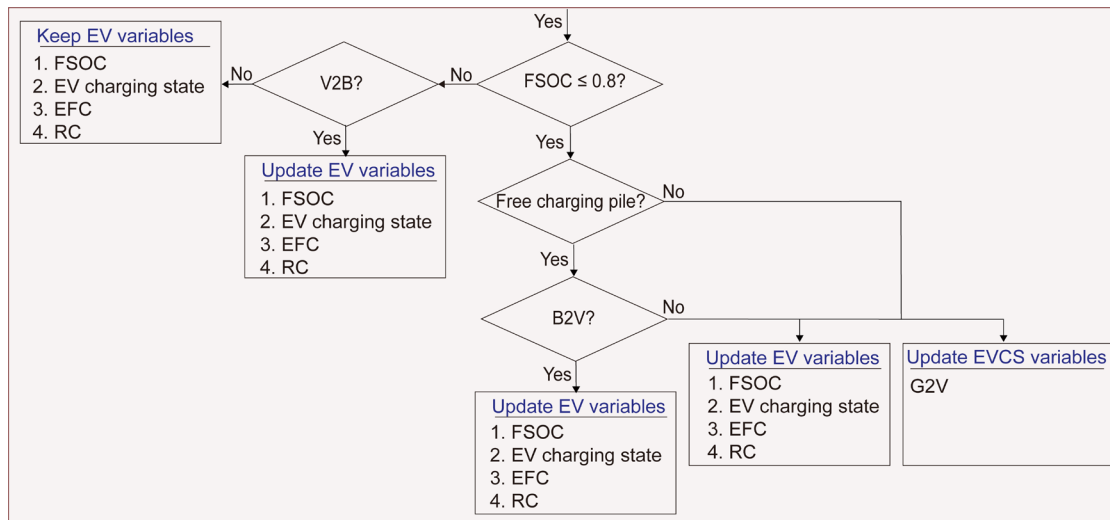


Figure S8 Methodology flow chart (EV at home).

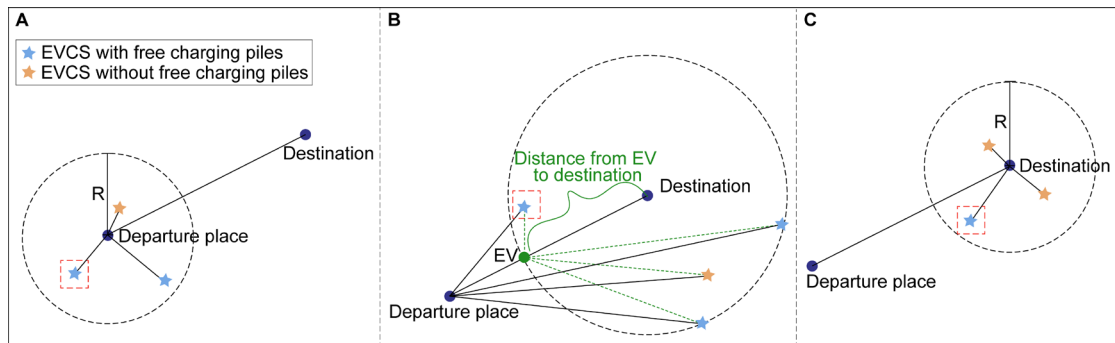


Figure S9 Electric vehicle charging station searching methods.

- (A) EVs at the departure place.
 (B) EVs at the road less than half of the trip.
 (C) EVs at the road near the destination.

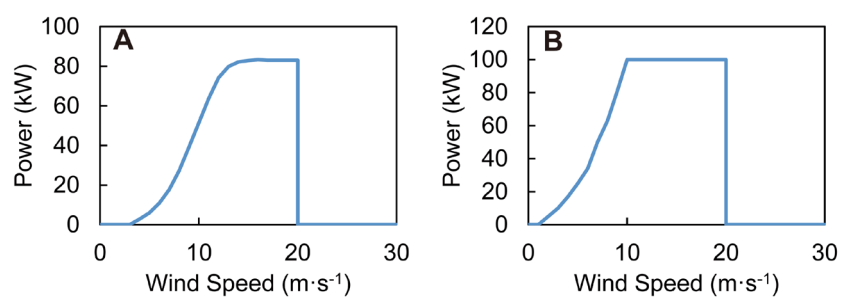


Figure S10 Power-wind speed curve of wind turbines.

(A) Hummer H21.0-60KW

(B) AN Bonus 100kW.^{1,2}

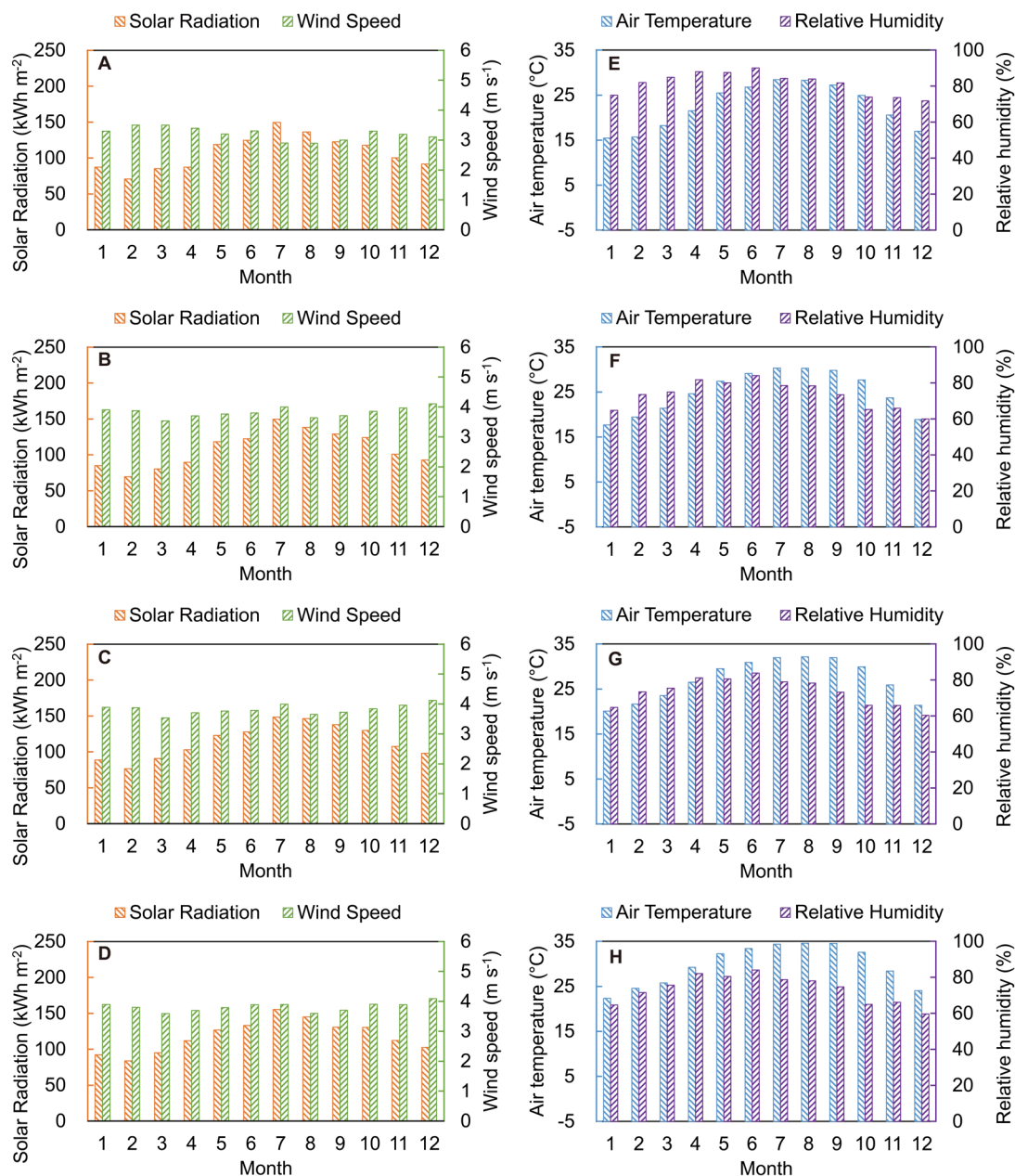


Figure S11 Weather data for Nansha District, Guangzhou.

- (A) The monthly solar radiation and monthly average wind speed in the typical year.
- (B) The monthly solar radiation and monthly average wind speed in 2020.
- (C) The monthly solar radiation and monthly average wind speed in 2060.
- (D) The monthly solar radiation and monthly average wind speed in 2100.
- (E) The monthly average air temperature and relative humidity in typical year.
- (F) The monthly average air temperature and relative humidity in 2020.
- (G) The monthly average air temperature and relative humidity in 2060.
- (H) The monthly average air temperature and relative humidity in 2100.

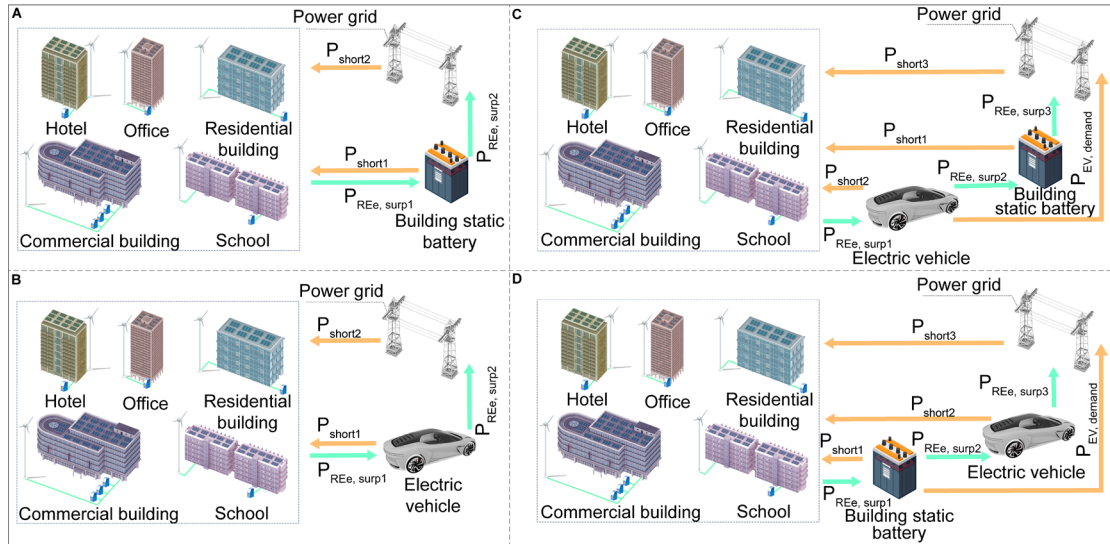


Figure S12 Building energy management.

(A) Building energy management with only energy storage batteries.

(B) Building energy interactive management with only electric vehicles.

(C) Building energy interaction management with electric vehicles and energy storage batteries, while electric vehicles have priority access to renewable energy.

(D) Building energy interaction management with electric vehicles and energy storage batteries, while energy storage batteries have priority access to renewable energy.

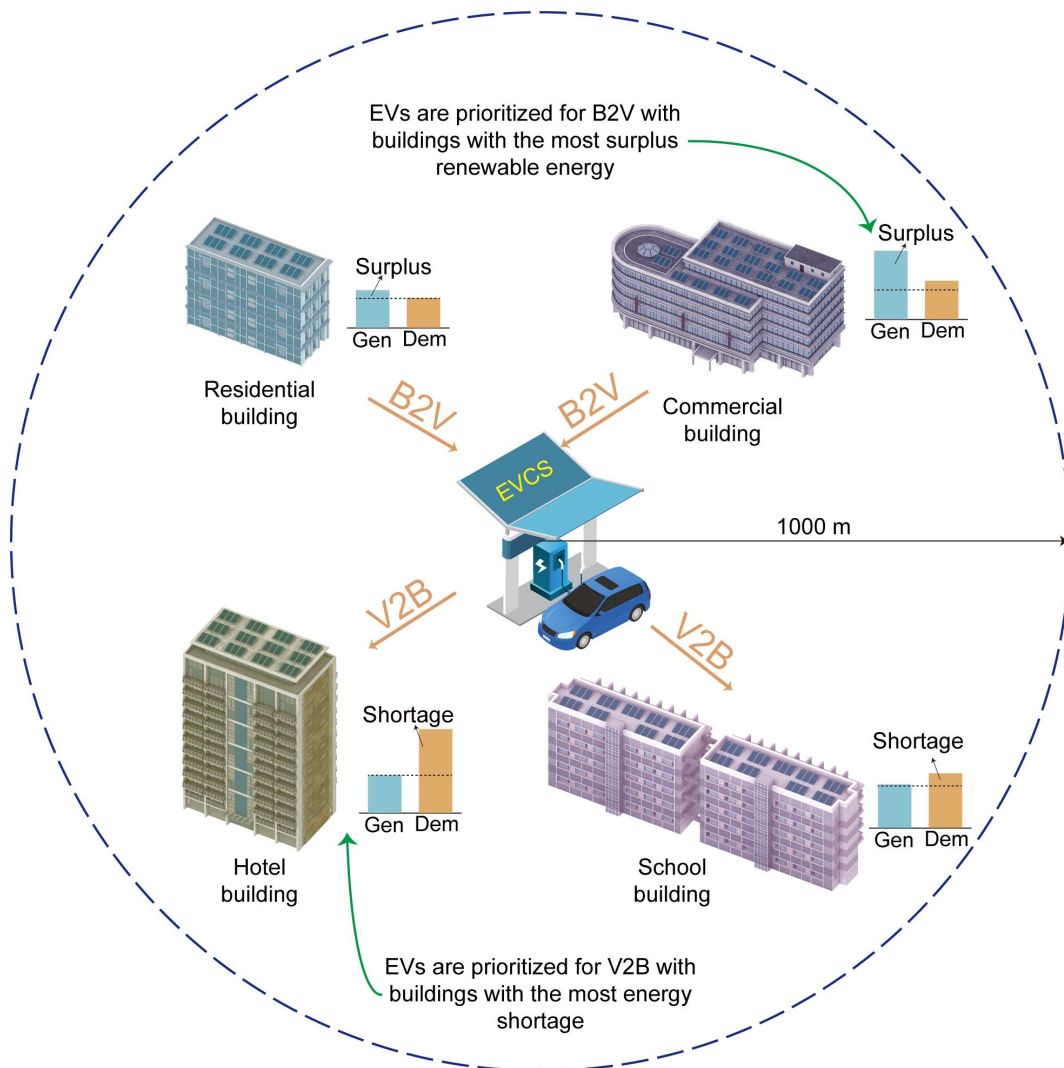


Figure S13 Charging station energy management.

Table S1 Peak powers of five typical TRNSYS models in the typical year, 2020, 2060, and 2100, respectively.

Typical year	commercial	301.74 kWh
	hotel	411.43 kWh
	office	497.58 kWh
	residential	36.82 kWh
	school	229.77 kWh
2020	commercial	307.06 kWh
	hotel	427.35 kWh
	office	565.57 kWh
	residential	35.74 kWh
	school	178.05 kWh
2060	commercial	317.27 kWh
	hotel	470.22 kWh
	office	605.64 kWh
	residential	41.02 kWh
	school	194.51 kWh
2100	commercial	331.50 kWh
	hotel	508.67 kWh
	office	687.14 kWh
	residential	43.09 kWh
	school	234.19 kWh

Table S2 Power limitations of coal-fired grid for five typical TRNSYS models.

Building type	Power limitations
commercial	319.60 kWh
hotel	469.77 kWh
office	611.31 kWh
residential	40.58 kWh
school	232.42 kWh

Table S4 The coefficient parameters of the $\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$ (NMC) battery capacity degradation curve⁴.

	$DoD = 1$	$DoD = 0.8$	$DoD = 0.65$	$DoD = 0.5$	$DoD = 0.35$	$DoD = 0.2$
k_1	-1.58E-11	-2.69E-11	-4.97E-11	-9.99E-11	-2.69E-10	-8.60E-10
k_2	3.66E-08	8.42E-08	1.37E-07	2.15E-07	4.34E-07	9.40E-07
k_3	-7.67E-05	-1.15E-04	-1.29E-04	-1.46E-04	-2.02E-04	-3.19E-04
k_4	1.00	1.00	1.00	1.00	1.00	1.00

Table S5 Probability distributions and descriptions of routes on weekdays.

Number of trips ^a	Route ^b	Probability ⁵	Types of destination buildings
Number of trips =2	1	27.9/58.8	Office
	2	10.3/58.8	School
	3	14.5/58.8	Commercial
	4	6.1/58.8	Hotel
Number of trips > 2	5	0.6	Office and commercial. Then, building types of destinations of the remaining trips follow the distributions of route1-route 4
	6	0.3	School and commercial. Then, building types of destinations of the remaining trips follow the distributions of route1-route 4
	7	0.1	All building types of destinations follow the distributions of route1-route 4

^a Trip: a trip represents a journey with a departure place and a destination.

^b Route: a route represents the combination of trips.

Table S6 Probability distributions and descriptions of routes on weekends.

Type of tour	Route	Probability ⁵	Type of destination buildings
Number of trips =2	1	16.6/55.1	Office
	2	9.7/55.1	School
	3	20.1/55.1	Commercial
	4	8.7/55.1	Hotel
Number of trips > 2	5	0.7	Office and commercial. Then, building types of destinations of the remaining trips follow the distributions of route1-route 4
	6	0.1	School and commercial. Then, building types of destinations of the remaining trips follow the distributions of route1-route 4
	7	0.2	All building types of destinations follow the distributions of route1-route 4

Table S7 Probability distribution of number of trips in a day.

Number of trips	2	3	4	5	6	7
Probability	0.52	0.37	0.08	0.02	0.005	0.005

Table S8 Electric vehicles (EVs) timetable on weekdays.

Type of tour	Type of destinations	Arrival time
Number of trips = 2	Commercial	[8:00, 19:00]
	Hotel	[8:00, 13:00]
	Office	8:00
	School	8:00
Number of trips > 2	Several destinations	[8:00, 14:00]
Go back home	Residence	[max ^a , 21:00]

^a max: max represents the arrival time of the EV's last trip before going back home plus one hour

Table S9 Electric vehicles (EVs) timetable on weekends.

Type of tour	Type of buildings	Arrival time
Number of trips = 2	Commercial	[8:00, 19:00]
	Hotel	[8:00, 13:00]
	Office	[8:00, 19:00]
	School	[8:00, 19:00]
Number of trips > 2	Several destinations	[8:00, 14:00]
Go back home	Residence	[max ^a , 21:00]

^a max: max represents the arrival time of the EV's last trip before going back home plus one hour

Table S10 Electric vehicles (EVs) speed at different time intervals.

Time interval	Average speed range (km/h)
4:00 A.M. - 7:00 A.M.	[40, 60]
7:00 A.M. - 10:00 A.M.	[20, 30]
10:00 A.M. - 17:00 P.M.	[30, 40]
17:00 P.M. - 23:00 P.M.	[20, 30]
23:00 P.M. - 4:00 A.M.	[40, 60]

Table S11 Parameters of electric vehicles (EVs).

Parameters	Information and value		
Car model	Aion S ⁶	Aion Y ⁷	Tesla Model Y ⁸
Battery type	NMC	LFP	LFP
Battery capacity (kWh)	58.8	80	81
Range (km)	510	600	488
Charging power	0.5C	0.5C	0.5C
Charging efficiency	0.95	0.95	0.95
Discharging efficiency	0.95	0.95	0.95

Table S12 Charging and discharging parameters of the piles.

Parameters	Information and value		
Charging pile power	7 kW ⁹	24 kW ¹⁰	90 kW ¹¹
Charging efficiency ^a	92%	90%	85%
Discharging efficiency ^b	92%	90%	85%

^a Charging efficiency: the electricity transfer efficiency of G2V or B2V process

^b Discharging efficiency: the electricity transfer efficiency of V2B process

Table S13 Description on different types of buildings.

Building type	Total building area (m ²)	Average number of floors	BIPV capacity (kWp)	Roof-top PV capacity (kWp)	On-site Wind turbine capacity (kW)
Commercial	1.994×10 ⁶	3.450	97,024	55,173	218,700
Hotel	7.430×10 ⁵	4.369	46,807	16,233	71,820
Office	7.980×10 ⁶	3.661	359,954	208,062	504,000
Residence	1.559×10 ⁷	9.081	972,037	163,924	637,200
School	1.694×10 ⁶	3.564	74,019	45,379	197,100

Table S14 The electrical performance of the installed roof-top PV module.

Electrical performance parameters (STC ^a)			
Maximum power ($P_{\max}$)	100.8 W	Operation temperature	-40~85 °C
Voltage at maximum power (V_{mp})	17 V	Size	0.89 m ²
Current at maximum power (I_{mp})	5.9 A	Temperature coefficient of I_{sc}	0.02 A/K
Open-circuit voltage (V_{oc})	21.6 V	Temperature coefficient (V_{oc})	-0.079 V/K
Short-circuit current (I_{sc})	6.5 A	Efficiency	0.1133%

^a Standard testing conditions (STC): Solar radiation at 1,000 W m⁻², Cell temperature at 25 °C

Table S15 The electrical performance of the installed BIPV module.

Parameters	Value
Cover emissivity	0.9
Cover conductivity (kJ/hr·m·K)	3.6
Cover thickness (m)	0.004
Substrate resistance (h·m ² ·K/kJ)	0.147
Channel emissivity - top	0.09
Channel emissivity - bottom	0.9
Back resistance (h·m ² ·K/kJ)	0.146
Channel height (m)	0.0508
Absorptance	0.9
Refractive index	1.526
Extinction coefficient (m ⁻¹)	4
Reference PV efficiency	0.1427
Reference temperature (K)	298
Reference radiation (kJ/hr·m ²)	3600
Efficiency modifier – temperature (1/K)	-0.000663
Efficiency modifier – radiation (h·m ² /kJ)	0.00006175

Table S16 Parameters of wind turbines.

Parameters	Information and Value	
	Hummer H21.0-60KW ¹	AN Bonus 100kW ²
Wind turbine		
Rated power (kW)	60	100
Cut-in wind speed (m·s ⁻¹)	2.0	2.0
Rated speed (m·s ⁻¹)	9.0	10.0
Cut-out wind speed (m·s ⁻¹)	20.0	20.0
Hub height (m)	21.0	30.0
Rotor diameter (m)	10.0	25.0

Table S17 Charging and discharging parameters of static batteries.

Parameters	Value
Charging limit	FSOC ^a < 0.9
Discharging limit	FSOC > 0.2
Max charging power	0.5 C
Max discharging power	0.5 C
Charging efficiency	95%
Discharging efficiency	95%

^a FSOC: fractional state of charge

Table S18 Charging and discharging parameters of electric vehicle (EV) batteries.

Parameters	Value
Charging limit	FSOC < 0.9
Discharging limit	FSOC > 0.2
Maximum charging power	0.5 C
Maximum discharging power to the building	0.5 C
Driving discharge coefficient	(0.9 - 0.2) / range ^a
Charging efficiency	95%
Discharging efficiency	95%

^a range: range refers to the distance an EV can travel on a full charge.

Table S19 Electricity price for residential buildings and school buildings in Guangzhou.

Summer ^a electricity price (CN¥ ^b kWh ⁻¹)				Non-summer ^c electricity price (CN¥ kWh ⁻¹)			
Energy imported per household per month (kWh)	Peak ^d	Flat ^e	Valley ^f	Energy imported per household per month (kWh)	Peak ^d	Flat ^e	Valley ^f
≤ 260	0.9950	0.5889	0.2292	≤ 200	0.9950	0.5889	0.2292
> 260 and ≤ 600	1.0450	0.6389	0.2792	> 200 and ≤ 400	1.0450	0.6389	0.2792
> 600	1.2950	0.8889	0.5292	> 400	1.2950	0.8889	0.5292

^a The summer months include May to October.

^b 1 CN¥=0.1375 US\$

^c The non-summer months include November to April.

^d Peak period is from 10:00 AM to 12:00 AM and from 14:00 PM to 19:00 PM.

^e Flat period is from 8:00 AM to 10:00 AM, from 12:00 PM to 14:00 PM and from 19:00 PM to 0:00 AM.

^f Valley period is from 0:00 AM to 8:00 AM.

Table S20 Electricity price for commercial users in Guangzhou.

Period	Super peak ^b	Peak ^c	Flat ^d	Valley ^e
Electricity price (CN¥ ^a kWh ⁻¹)	1.7156	1.3780	0.8220	0.3296

^a 1 CN¥=0.1375 US\$

^b Super peak period is from 11:00 AM to 12:00 AM and from 15:00 PM to 17:00 PM in July, August, and September.

^c Peak period is from 10:00 AM to 11:00 AM, from 14:00 PM to 15:00 PM and from 17:00 PM to 19:00 PM in July, August, and September. In other periods (except for July, August, and September), the peak period is from 10:00 AM to 12:00 AM and from 14:00 PM to 19:00 PM.

^d Flat period is from 8:00 AM to 10:00 AM, from 12:00 AM to 14:00 PM and from 19:00 PM to 0:00 AM.

^e Valley period is from 0:00 AM to 8:00 AM.

Table S21 Unit price of initial purchase and maintenance of renewable equipment and charging piles.

Equipment	Capital price (US\$)	Maintenance price (US\$/a)
PV panel (kWp ⁻¹)	240 ¹²	12 ¹³
Wind turbine (kW ⁻¹)	400 ¹⁴	20 ¹³
LFP battery (kWh ⁻¹)	118 ¹⁵	0.059
NMC battery (kWh ⁻¹)	173 ¹⁵	0.865
Charging pile (7 kW)	500	10
Charging pile (24 kW)	2,142.86	42.85
Charging pile (90 kW)	5,000	100

Note S1 Power outage hours simulation based on TRNSYS models.

As detailed in the section on **District building energy demand and generation modelling**, the method for simulating hourly energy demand of urban buildings involves several steps. First, five typical building models are created using TRNSYS 18 software, which outputs the hourly energy demand for each model. This demand is then scaled proportionally based on the gross building area to determine the hourly energy demand for each building in the city.

A similar approach is applied to establish the coal-fired grid power limitation for each building. We assume that with global warming, heatwaves become frequent and prolonged. When a building's instantaneous power demand exceeds its coal-fired grid limitation, a power outage occurs during that hour. Using peak demand data from five typical buildings under typical, 2020, 2060, and 2100 weather conditions, as shown in **Table S1**, the 60th percentile within the minimum and maximum peak demand power is used as the building's coal-fired grid power limitation, as shown in **Table S2**. This percentile is chosen because it reflects few power outage hours in and before 2020 while anticipating increased outage hours in 2060 and 2100 due to global warming. Therefore, it is allowed to compare the impact of climate-adaptive strategy on urban energy resilience against the reference strategy.

Note S2 Detailed information of TRNSYS building models.

There are five types of buildings in Nansha District: commercial buildings, hotel buildings, office buildings, residential buildings and school buildings. The energy consumption of these buildings is simulated by TRNSYS model¹⁶. The TRNSYS model description of these five types of buildings is as follows:

The commercial buildings in the TRNSYS model have two floors, each floor is 2640 m² (66 m × 40 m) and 3.2 meters high. The sources of internal gains of commercial buildings are equipment, light and human. Internal gains from equipment are 13 W m⁻², internal gains from light are 16 W m⁻², and internal gains generated by the human body are 126 W per person. The building saturation population is 2,376. In the HVAC system, the cooling system set point temperature is 24 °C with an air humidity at 55%, which are cooled by air handling unit (AHU) cooling system and space cooling system respectively. The cooling source is two Carrier30RB302 chillers with a rated power of 293 kW. The commercial building does not have a hot water system because its hot water demand is 0. In terms of renewable energy systems, the commercial building has 2,500 roof-top PV panels (**Table S14**), BIPVs on the walls (**Table S15**), and two on-site wind turbines with 60 kW and 100 kW respectively (**Table S16**).

The hotel building in the TRNSYS model has 30 floors, each floor is 400 m² (20 m × 20 m) and 3.2 meters high. The sources of internal gains of hotel buildings are equipment, light and human. Internal gains from equipment are 4 W m⁻², internal gains from light are 17 W m⁻², and internal gains generated by the human body are 126 W per person. The building saturation population is 480. In the HVAC system, the cooling system set point temperature is 24 °C with an air humidity at 55%, which are cooled by air handling unit (AHU) cooling system and space cooling system respectively. The cooling source is a TRANE CGAF C40 chiller with a rated power of 103 kW and a TRANE R22 080 chiller with a rated power of 236 kW. The daily hot water consumption at 60 °C for the hotel building is 1.1 m³ (floor·day)⁻¹. In terms of renewable energy systems, the hotel building has 400 roof-top PV panels (**Table S14**), BIPVs on the walls (**Table S15**), and two on-site wind turbines with 60 kW and 100 kW respectively (**Table S16**).

The office building in the TRNSYS model has 30 floors, each floor is 400 m² (20 m × 20 m) and 3.2 meters high. The sources of internal gains of office buildings are equipment, light and human. Internal gains from equipment are 10 W m⁻², internal gains from light are 15 W m⁻², and internal gains generated by the human body are 126 W per person. The building saturation population is 900. In the HVAC system, the cooling system set point temperature is 24 °C with an air humidity at 55%, which are cooled by air handling unit (AHU) cooling system and space cooling system respectively. The cooling source is a Midea 250W chiller with a rated power of 250 kW and a TRANE R407C 120 chiller with a rated power of 400 kW. The daily hot water consumption at 60 °C for the office building is 0.13 m³ (floor·day)⁻¹. In terms of renewable energy systems, the office building has 400 roof-top PV panels (**Table S14**), BIPVs on the walls (**Table S15**), and two on-site wind turbines with 60 kW and 100 kW respectively (**Table S16**).

The residential building in the TRNSYS model has 3 floors, each floor is 400 m² (20 m × 20 m) and 3.2 meters high. The sources of internal gains of residential buildings are equipment, light and human. Internal gains from equipment are 4 W m⁻², internal gains from light are 17 W m⁻², and internal gains generated by the human body are 126 W per person. The building saturation population is 48. In the HVAC system, the cooling system set point temperature is 24 °C with an air humidity at 55%, which are cooled by air handling unit (AHU) cooling system and space cooling system respectively. The cooling source is a Midea MGC V12WD2RN1 chiller with a rated power of 11.2 kW and a TRANE 30RH 027 chiller with a rated power of 26.3 kW. The daily hot water consumption at 60 °C for the residential building is 1.1 m³ (floor·day)⁻¹. In terms of renewable energy systems, the residential building has 400 roof-top PV panels (**Table S14**), BIPVs on the walls (**Table S15**), and two on-site wind turbines with 60 kW and 100 kW respectively (**Table S16**).

The school building in the TRNSYS model has 5 floors, each floor is 900 m² (30 m × 30 m) and 4.0 meters high. The sources of internal gains of school buildings are equipment, light and human. Internal gains from equipment are 10 W m⁻², internal gains from light are 10 W m⁻², and internal gains generated by the human body are 126 W per person. The building saturation population is 500. In the HVAC system, the cooling system set point temperature is 24 °C with an air humidity at 55%, which are cooled by air handling unit (AHU) cooling system and space cooling system respectively. The cooling source is two Carrier30RB192 chillers with a rated power of 193 kW. The daily hot water consumption at 60 °C for the school building is 0.5 m³ (floor·day)⁻¹. In terms of renewable energy systems, the school building has 900 roof-top PV panels (**Table S14**), BIPVs on the walls (**Table S15**), and two on-site wind turbines with 60 kW

and 100 kW respectively (**Table S16**).

Note S3 Renewable energy simulation process of TRNSYS models.

In this study, three types of on-site renewable systems are implemented, i.e., BIPVs, roof-top PVs and wind turbines. Vertical BIPVs are designed on building façades (windows are excluded) (Note: The window-to-wall ratio of all buildings is set to 0.2). The monocrystalline silicon PVs are adopted with detailed parameters as listed in **Table S15**. In TRNSYS 18, Type 567 is used for the dynamic simulation of BIPVs.

The rooftop photovoltaic (PV) system design occupies 90% of the building's roof area. The parameters for the PV panels used are listed in **Table S14**. In TRNSYS18, Type103b is employed to simulate the dynamic performance of the PV system.

On-site wind turbines are set to provide renewable electricity based on wind speed in Nansha District, Guangzhou. Each building in Nansha District is equipped with a 60 kW and a 100-kW wind turbine. **Table S16** and **Figure S10** list the parameters of wind turbines.

There are two reasons for equipping each building with a 60 kW and a 100 kW wind turbine. First, the U.S. Department of Energy recommends installing only small wind turbines for individual buildings, with sizes ranging from 20 W to 100 kW¹⁷. Second, our calculations indicate that equipping each building in the Nansha District with a 60 kW and a 100 kW wind turbine, in conjunction with Building Integrated Photovoltaics (BIPV) and rooftop PV systems, can achieve near net-zero energy. This approach promotes sustainable development, which is a key direction for the future of building and urban development.

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